

PROFESSOR: TILMAN BÖRGERS

MICROECONOMICS 601
UNIVERSITY OF
MICHIGAN

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Class 1: Consumer Choice, Preferences, and Rationalization

From Utility Maximization to Choice Correspondences

A consumer chooses a bundle $x = (x_1, \dots, x_\ell) \in \mathbb{R}_+^\ell$ to

$$\max_{x \in \mathbb{R}_+^\ell} u(x) \quad \text{s.t.} \quad p \cdot x \leq y,$$

where $p \in \mathbb{R}_{++}^\ell$ is the price vector and $y > 0$ income.

Definition (Ordinal utility & monotone transforms). A function $u : X \rightarrow \mathbb{R}$ represents a preference relation $\succeq$ on X if

$$x \succeq y \iff u(x) \geq u(y) \quad \text{for all } x, y \in X.$$

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing, then $v = f \circ u$ represents the same $\succeq$. Hence utility is *ordinal*: only the induced ranking matters.

Example (Log-transform of Cobb–Douglas). If $u(x_1, x_2) = x_1 x_2$, then $v(x_1, x_2) = \ln x_1 + \ln x_2 = f(u)$ with $f(t) = \ln t$ (strictly increasing on $t > 0$) represents the same preferences.

Definition (Feasible sets and choice). Let X be the set of feasible objects and $\mathcal{A}$ a family of nonempty subsets $A \subseteq X$ (“menus”). A *choice correspondence* is a mapping

$$c : \mathcal{A} \rightrightarrows 2^X, \quad c(A) \subseteq A,$$

assigning to each menu the set of chosen elements.

“No off-menu”: the condition $c(A) \subseteq A$ forbids choosing something that was not available.

Quick read and intuition. X is the *universe* of feasible objects (all alternatives that could be considered). A *menu* $A \subseteq X$ is the subset actually on the table in a decision instance (what is “on offer”). The choice correspondence c takes each menu A and returns the set of alternatives actually chosen when A is offered. The double arrow $c : \mathcal{A} \rightrightarrows X$ reminds us that c may be *multi-valued*.

WHY A *correspondence* AND NOT A FUNCTION?

There are two main reasons:

- **Tie/indifference:** if the person is indifferent between x and y in A , the model does not impose a tie-break, so $c(A)$ may include both.
- **Repeated data:** if we observe several decisions with the same A and sometimes x is chosen and other times y , it is natural to record $c(A) = \{x, y\}$.

If there is always a unique chosen element, c is called a choice *function*.

Link to utility. If there exists a preference $\succeq$ (or a utility u that represents it), a standard way to rationalize c is

$$c(A) = \arg \max_{x \in A} u(x).$$

Ties in $u \Rightarrow c(A)$ may contain multiple elements. With strictly convex preferences over convex sets (e.g., $\mathbb{R}_+^\ell$ with Cobb–Douglas) the arg max is unique; with discrete domains or preferences with flat segments, multiple optima may appear.

Domain of c . $\mathcal{A}$ is the family of menus on which we define/observe choice. In revealed-preference theory we often take *all* nonempty subsets of X ; with real data, $\mathcal{A}$ consists only of the menus that were observed.

Non-emptiness is often imposed: $c(A) \neq \emptyset$ for every $A \in \mathcal{A}$. In finite domains this is reasonable; in infinite ones it requires continuity/compactness assumptions for a maximum to exist.

Example: Discrete menus and budget sets.

Discrete. Let $X = \{\text{Air}, \text{ThinkPad}, \text{XPS}\}$ and suppose we observe:

$$\begin{aligned} c(\{\text{Air}, \text{ThinkPad}\}) &= \{\text{Air}, \text{ThinkPad}\} && \text{(sometimes Air is chosen, sometimes ThinkPad),} \\ c(\{\text{ThinkPad}, \text{XPS}\}) &= \{\text{ThinkPad}\}, \\ c(\{\text{Air}, \text{XPS}\}) &= \{\text{Air}\}. \end{aligned}$$

Here c is a *correspondence* because for the first menu there is a tie and we record both choices. If later the menu is $\{\text{Air}, \text{ThinkPad}, \text{XPS}\}$, coherence (WARP) requires not choosing ThinkPad while Air is available if Air was ever chosen over ThinkPad.

Budget set: Let $X = \mathbb{R}_+^2$ and $A = B(p, y) = \{x \in \mathbb{R}_+^2 : p_1 x_1 + p_2 x_2 \leq y\}$.

- Cobb–Douglas. For $u(x) = x_1^\alpha x_2^{1-\alpha}$ with $\alpha \in (0, 1)$, the FOCs yield

$$x_1^* = \frac{\alpha}{p_1} y, \quad x_2^* = \frac{1-\alpha}{p_2} y.$$

Hence $c(A) = \{(x_1^*, x_2^*)\}$ is *unique*. Strict convexity of preferences guarantees no ties.

- Perfect complements (Leontief). For $u(x) = \min\{x_1, x_2\}$ the optimum occurs at the kink $x_1 = x_2$ (consumption in 1:1 proportion) subject to the budget:

$$x_1^* = x_2^* = \frac{y}{p_1 + p_2}.$$

Here too $c(A)$ is *unique*: the indifference “L” has a single intersection with the budget line when $p_1, p_2 > 0$.

- Perfect substitutes. For $u(x) = x_1 + x_2$:

$$c(A) = \begin{cases} \{(y/p_1, 0)\}, & \text{if } p_1 < p_2 \quad (\text{corner at } x_1), \\ \{(0, y/p_2)\}, & \text{if } p_2 < p_1 \quad (\text{corner at } x_2), \\ \{x \in \mathbb{R}_+^2 : p_1 x_1 + p_2 x_2 = y\}, & \text{if } p_1 = p_2 \quad (\text{all points on the segment}). \end{cases}$$

This last case illustrates why we model c as a *correspondence*: when the slope of the budget line matches that of indifference curves, there are *infinitely many* optima.

In continuous domains $c(A)$ can be a singleton (strictly convex preferences or “point” kinks) or multivalued (flat segments or slope coincidences). In discrete data, c becomes multivalued when we observe ties or different choices under the same menu.

Preferences and Their Basic Properties

Definition (Preference primitives). A *weak* preference $\succeq$ is a binary relation on X . Define the *strict* preference $x \succ y$ iff $x \succeq y$ and not $y \succeq x$, and *indifference* $x \sim y$ iff $x \succeq y$ and $y \succeq x$.

Definition (Rational (standard) axioms). A preference $\succeq$ is *complete* if for all $x, y \in X$ we have $x \succeq y$ or $y \succeq x$ (or both); and *transitive* if $x \succeq y$ and $y \succeq z$ imply $x \succeq z$.

Remark. Completeness formalizes “the agent can compare any two options”; transitivity formalizes *consistency* of those comparisons. With these, the derived $\sim$ is an equivalence relation.

From Choice Data to Preferences: Coherence (WARP)

Definition (Non-emptiness). c satisfies *non-emptiness* if $c(A) \neq \emptyset$ for every $A \in \mathcal{A}$.

Definition (Choice coherence = WARP). c satisfies *choice coherence* (WARP) if for all menus $A, B \in \mathcal{A}$ and all $x, y \in A \cap B$:

$$(x \in c(A) \text{ and } y \in A \setminus c(A)) \implies y \notin c(B).$$

Interpretation: If x is chosen over y when both were available, then y is never chosen in any other menu that still contains x .

Definition (Rationalization by a preference). A preference $\succeq$ *rationalizes* c if for every $A \in \mathcal{A}$,

$$c(A) = \{x \in A : x \succeq y \text{ for all } y \in A\}.$$

That is, $c(A)$ is the set of $\succeq$ -maximal elements in A .

Key Equivalences on Finite Domains

Throughout this section assume X is finite (as in our class; Kreps develops the infinite case).

Theorem 1 (Revealed preference characterization on finite X). *Let X be finite and c defined on all nonempty $A \subseteq X$. Then the following are equivalent:*

1. c satisfies non-emptiness and choice coherence (WARP).
2. There exists a complete and transitive preference $\succeq$ that rationalizes c (i.e., $c(A)$ are $\succeq$ -maxima in A for all A).

Proof sketch your professor outlined. (2) $\Rightarrow$ (1): If $c(A)$ are the $\succeq$ -maximal elements in each A , then $c(A) \neq \emptyset$ (finite sets have maxima under complete, transitive $\succeq$), and WARP holds because if $x \succeq y$ in A with $y \notin c(A)$, then in any B containing both, y cannot beat x .

(1) $\Rightarrow$ (2) (construction): Define a *revealed weak preference* by

$$x \succeq^* y \iff x \in c(\{x, y\}).$$

Completeness: Because $c(\{x, y\}) \neq \emptyset$ for every pair, either x or y is chosen from $\{x, y\}$, hence either $x \succeq^* y$ or $y \succeq^* x$.

Transitivity: Suppose $x \succeq^* y$ and $y \succeq^* z$. If transitivity failed ($x \not\succeq^* z$), then $z \in c(\{x, z\})$. Consider menu $A = \{x, y, z\}$. WARP applied to the pairs $\{x, y\}$ and A forces $y \notin c(A)$; WARP applied to $\{y, z\}$ and A forces $z \notin c(A)$. But then $c(A)$ would be empty, contradicting non-emptiness. Hence $x \succeq^* z$.

Finally, verify that $c(A)$ equals the set of $\succeq^*$ -maximal elements of A (otherwise WARP would be violated by the offending pair inside A). $\square$

Corollary (Utility representation on finite X). *If $\succeq$ is complete and transitive on finite X , then there exists a utility $u : X \rightarrow \mathbb{R}$ that represents $\succeq$. For instance, assign u by ranks: largest u to any $\succeq$ -maximal elements, then proceed recursively on the remainder. Any strictly increasing transform $f \circ u$ represents the same $\succeq$.*

This answers your "Why?": Non-emptiness on all pairs forces a choice between any two options, which is exactly completeness of $\succeq^*$.

Linking Back to Utility Maximization

Proposition (Utility maximization rationalizes c). Fix a preference $\succeq$ represented by u . Define $c(A) = \arg \max_{x \in A} u(x)$. Then c satisfies non-emptiness and WARP, and c is rationalized by $\succeq$.

Remark (When do we need more structure (continuity, convexity)?). For infinite X (e.g., $X = \mathbb{R}_+^\ell$), completeness and transitivity alone do not guarantee a continuous or nice u . Standard micro adds continuity (to rule out jumps), monotonicity (“more is better”), and sometimes (strict) convexity (diminishing marginal rate of substitution). In the finite case of this class, Theorem 1 and the ranking construction suffice.

Quick Checks & Exercises

Exercise 0.0.1 (Diagnose a coherence (WARP) violation). Suppose $c(\{x, y\}) = \{x\}$ and $c(\{x, y, z\}) = \{y\}$. Show this violates WARP. Which pair produces the contradiction?

Exercise 0.0.2 (Reconstructing $\succeq$ from c). Given choice data on all nonempty $A \subseteq X$ for finite X , build $\succeq$ as in the proof and show $c(A)$ equals the set of $\succeq$ -maximal elements of A for every A .

Exercise 0.0.3 (Monotone transforms). Let u represent $\succeq$ on $X \subseteq \mathbb{R}_+^\ell$. Prove that $v = f \circ u$ represents $\succeq$ if and only if f is strictly increasing on $u(X)$.

Exercise 0.0.4 (Utility from any complete, transitive $\succeq$ on finite X). Construct a representing u by top-cycling ranks: set $u(x) = k$ for all x in the k -th indifference “layer” in the order from best to worst. Argue that this works and is unique up to strictly increasing transforms.

Mini FAQ

Q: Why does non-emptiness on all pairs imply completeness? Because for each $\{x, y\}$ the choice set $c(\{x, y\})$ contains either x or y . Declare $x \succeq y$ iff $x \in c(\{x, y\})$. Then for every pair, at least one direction holds—exactly completeness.

Q: Is choice coherence the same as WARP? Yes in this finite, menu-based setting: “If x was chosen over y somewhere, never choose y when x is also available.”

Class 2: Rationality of Choice (finite and infinite)

Quick recap (finite case)

Let X be the set of alternatives (finite in this subsection) and let $\mathcal{A} \subseteq 2^X \setminus \{\emptyset\}$ be the family of feasible menus.

Definition (Choice function). A *choice function* is a mapping $c : \mathcal{A} \Rightarrow 2^X$ such that $c(A) \subseteq A$ for every $A \in \mathcal{A}$. We say that c satisfies *nonemptiness* if $c(A) \neq \emptyset$ for all $A \in \mathcal{A}$.

Definition (Preferences and utility). A *preference relation* is a complete and transitive binary relation $\succeq \subseteq X \times X$. Write $x \succ y$ for strict preference and $x \sim y$ for indifference. A *utility function* is a map $u : X \rightarrow \mathbb{R}$.

Definition (Rationalization and representation). We say that $\succeq$ *rationalizes* c if, for every $A \in \mathcal{A}$,

$$c(A) = \{x \in A : x \succeq y \ \forall y \in A\}.$$

We say that u *represents* $\succeq$ if $u(x) \geq u(y) \iff x \succeq y$ for all $x, y \in X$.

Consistency of choice (WARP). We use the operational form

(WARP) If $x, y \in A \cap B$, $x \in c(A)$ and $y \in c(B)$, then $x = y$.

Equivalently: if $x, y \in A$, $x \in c(A)$, $y \notin c(A)$, and $x \in B$, then $y \notin c(B)$.¹

Proposition (Rationalization on finite X). *There exists a complete and transitive preference relation $\succeq$ that rationalizes the choice function c if and only if the choice function c satisfies nonemptiness and WARP.*

Proof idea. ($\Rightarrow$) If c comes from maximizing $\succeq$ in each menu, a maximal element always exists (nonemptiness), and no “revenge” by y over x is possible when $x \succeq y$ (WARP).

($\Leftarrow$) Build $\succeq$ from pairwise choices:

$$x \succeq y \iff x \in c(\{x, y\}).$$

¹ *Intuition for future me:* once x has been chosen over y when both were available, y cannot “beat” x in any other menu where both remain present.

We want, for all A ,

$$x \in c(A) \iff x \succeq y \quad \forall y \in A.$$

($\Rightarrow$) By *contraposition*: if $x \in c(A)$ but $\exists y \in A$ with $x \notin c(\{x, y\})$, then $c(\{x, y\}) = \{y\}$. WARP forbids x being chosen in A with y available—contradiction.

($\Leftarrow$) If $x \succeq y$ for all $y \in A$, then x wins every pairwise contest in A . If $x \notin c(A)$, nonemptiness gives some $y \in c(A)$. But WARP, applied to $\{x, y\}$ and A , forbids choosing y in A when x was available and also wins the binary comparison—contradiction. $\square$

Proposition (Utility on finite X). *If $\succeq$ is complete and transitive on finite X , there exists $u : X \rightarrow \mathbb{R}$ representing $\succeq$. Construction: rank X and assign increasing values (e.g., $u = 3, 2, 1, \dots$).*

Proposition (Invariance to monotone transformations). *If u represents $\succeq$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing on the range of u , then $f \circ u$ also represents $\succeq$.*

“step-by-step” proof of Proposition

Define the preference relation $\succeq$ via binary choices: $x \succeq y \iff x \in c(\{x, y\})$. We seek, for all A :

$$x \in c(A) \iff x \in c(\{x, y\}) \quad \forall y \in A.$$

(i) $\Rightarrow$ If $x \in c(A)$ but $\exists y \in A$ with $x \notin c(\{x, y\})$, then $c(\{x, y\}) = \{y\}$. By WARP, y cannot be chosen in any menu where x is present. Since $y \in A$ and $x \in A$, this contradicts $x \in c(A)$.

(ii) $\Leftarrow$ Suppose $x \notin c(A)$. By nonemptiness, $\exists y \in c(A)$. Since $x \succeq y$ by hypothesis (it wins every binary), we have $x \in c(\{x, y\})$. Applying WARP between $\{x, y\}$ and A , y could not be chosen in A . Contradiction.

Intuition comment. Part (ii) formalizes: “if x defeats everyone head-to-head, it must be the tournament champion.”²

² *Intuition for future me:* with finite X , WARP kills cycles and ensures the *binary maximum* is a menu maximum.

What if X is infinite? Why things can fail without extra axioms

Example. Let $X = \mathbb{R}_+^2$ and $u(x_1, x_2) = x_1 + x_2$. On the “unrestricted” menu $A = X$, there is no maximizer: $c(X) = \emptyset$ (you can always increase a bit more).

To recover a useful notion of rationality on infinite sets, we impose three minimal conditions on c :

(i) *Finite nonemptiness:* if A is finite and nonempty, then $c(A) \neq \emptyset$.

(ii) *Choice coherency (WARP)* .

(iii) *Binary consistency (pairwise maximality)*: if $x \in A$ and $x \in c(\{x, y\})$ for every $y \in A$, then $x \in c(A)$.

Proposition (Rationalization on infinite X). *If c satisfies (i)–(iii), then there exists a complete and transitive $\succeq$ that rationalizes c (constructed from binary choices as above).*

Idea. (i)–(ii) ensure the induced binary relation is consistent; (iii) promotes the “pairwise champion” to menu champion even when A is infinite.³

Proposition (Utility representation on infinite X (countable order-dense criterion)). *Let $\succeq$ be a weak order (complete and transitive) on X . There exists $u : X \rightarrow \mathbb{R}$ that represents $\succeq$ iff there is a countable subset $X^* \subseteq X$ order-dense such that*

$$\forall x \succ y \quad \exists x^* \in X^* \text{ with } x \succeq x^* \succ y.$$

Intuition. In $(\mathbb{R}, \geq)$, $\mathbb{Q}$ is countable and *dense*: if $x > y$, you can insert a rational q with $x \geq q > y$. That property lets us “number” the order with reals without big jumps and build u . Then Proposition ensures any strictly increasing transform preserves $\succeq$.⁴

³ *Intuition for future me:* (iii) is exactly the missing piece when you passed from pairs to large menus; it formalizes “if x wins locally against every rival, it must win globally.”

⁴ *Intuition for future me:* order-density is the substitute for finiteness: “I can always slip in a numeric marker between any two strictly ordered alternatives.”

Class 3: Consumer Choice and Demand (Kreps Ch. 2)

Primitives and the baseline assumptions

We work with k goods and the consumption set $X = \mathbb{R}_+^k$. A bundle is $x = (x_1, \dots, x_k) \in X$.

Definition (Preferences on X). A (weak) preference relation $\succeq \subseteq X \times X$ is assumed *complete and transitive*. Write

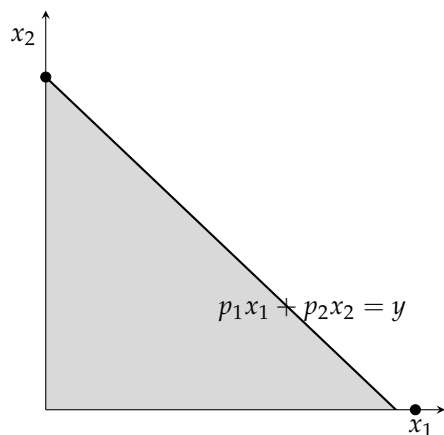
$$x \succ y \iff (x \succeq y \text{ and not } y \succeq x), \quad x \sim y \iff (x \succeq y \text{ and } y \succeq x).$$

Definition (Utility representation (not assumed a priori)). A function $u : X \rightarrow \mathbb{R}$ represents $\succeq$ if $u(x) \geq u(y) \iff x \succeq y$.

Budget sets and (Walrasian) demand

Let prices be $p \in \mathbb{R}_{++}^k$ and income $y > 0$. The budget set is

$$B(p, y) := \{x \in X : p \cdot x \leq y\}.$$



Intuition for future me. In this class, we **only** assume completeness and transitivity. In infinite domains, this *does not* guarantee maximizers or continuous utility representation. Several conclusions from the finite section do not automatically apply.

Intuition for future me. $d(p, y)$ is (again) a *correspondence*: it can be multivalued or even empty if there are no maxima (e.g., preferences without continuity/compactness). With $p \gg 0$, $B(p, y)$ is a bounded polytope in $\mathbb{R}_+^k$.

Figure 1: Budget set $B(p, y)$ in $k = 2$: region under the budget line.

Definition (Demand correspondence). The (Walrasian) demand at (p, y) is the set of $\succeq$ -maximal elements in the budget set:

$$d(p, y) := \{x \in B(p, y) : x \succeq x' \text{ for all } x' \in B(p, y)\}.$$

Indifference “curves” versus mere points

With additional structure, indifference sets typically look like smooth, downward-sloping, convex curves. Without such assumptions we cannot guarantee any shape—empirically we might only “see” scattered points.

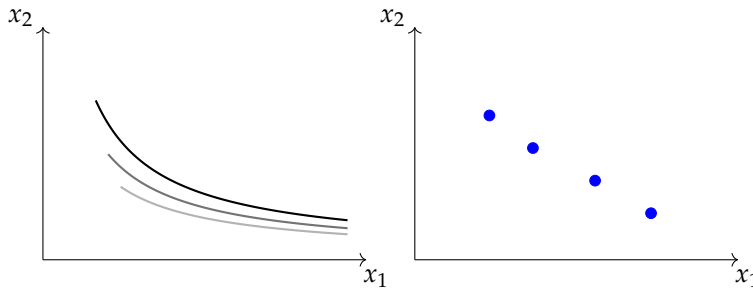


Figure 2: Left: “indifference curves” under regularity. Right: without assumptions, we may only observe a set of points.

Structural properties (preferences and utility)

Definition (Monotonicity). A preference $\succeq$ on $\mathbb{R}_+^k$ is *monotone* if for any $x, x' \in \mathbb{R}_+^k$,

$$x_i \geq x'_i \text{ for all } i = 1, \dots, k \implies x \succeq x'.$$

Equivalently, “more doesn’t hurt you” componentwise.

Definition (Strict/strong monotonicity). $\succeq$ is *strictly* (a.k.a. *strongly*) monotone if the above holds and, whenever $x_i \geq x'_i$ for all i and $x_j > x'_j$ for some j , then $x \succ x'$. More compactly:

$$x \geq x' \text{ and } x \neq x' \implies x \succ x'.$$

Remark. Together with mild continuity/convexity, monotonicity delivers well-behaved indifference sets and existence/uniqueness results for $d(p, y)$; we will call on them when needed below.

Monotonicity $\leftrightarrow$ (non)decreasing utility

Definition (Strict/strong monotonicity). A preference $\succeq$ is *strictly* (a.k.a. *strongly*) monotone if

$$x \geq x' \text{ and } x \neq x' \implies x \succ x'.$$

Intuition for future me. Indifference curves emerge from structural assumptions (continuity, monotonicity, convexity). Without them, the set $\{x : x \sim \bar{x}\}$ can be pathological or disconnected.

Intuition for future me. Under monotonicity and $p \gg 0$, any optimum $x^* \in d(p, y)$ must exhaust the budget: $p \cdot x^* = y$. Otherwise, you could increase a good a little bit and improve (or not worsen). ICs are non-increasing and the NE quadrant is “better”.

Proposition (Equivalence at the utility level). Assume $\succeq$ admits a utility representation (no continuity required).⁵

- (a) $\succeq$ is monotone $\iff$ every utility u representing $\succeq$ is nondecreasing in the product order:

$$x \geq x' \implies u(x) \geq u(x').$$

- (b) A preference relation $\succeq$ is strictly monotone $\iff$ every representing utility u is strictly increasing:

$$x \geq x', x \neq x' \implies u(x) > u(x').$$

Insatiability: global vs. local

Definition (Global insatiability (GI)). A preference relation $\succeq$ is *globally insatiable* if

$$\forall x \in X \exists x' \in X \text{ such that } x' \succ x.$$

Definition (Local insatiability (LI)). A preference relation $\succeq$ is *locally insatiable*⁶ if

$$\forall x \in X, \forall \varepsilon > 0 \exists x' \in X \text{ with } \|x' - x\| \leq \varepsilon \text{ and } x' \succ x.$$

Remark. $\text{LI} \Rightarrow \text{GI}$ (take, e.g., $\varepsilon = 1$). The converse fails: one may always find some improvement, but only far away.

Example (GI without monotonicity). On $X = \mathbb{R}_+^2$, $u(x_1, x_2) = x_1 - x_2$ yields GI: for any x , $(x_1 + 1, x_2) \succ x$. But preferences are not monotone (raising x_2 makes you worse off).

Remark (LI and budget exhaustion). If $p \gg 0$, $y > 0$ and $\succeq$ is LI, then any $x^* \in d(p, y)$ satisfies $p \cdot x^* = y$. *Idea:* if $p \cdot x^* < y$, LI yields x' arbitrarily close with $x' \succ x^*$; for small ε , $p \cdot x' \leq y$, contradicting optimality.

Convexity and upper contour sets

Definition (Convex preferences). A preference relation $\succeq$ is *convex* if for every $z \in X$ the weakly preferred set (same as the upper contour set at z)

$$U(z) := \{x \in X : x \succeq z\} = \text{NWT}(z)$$

is convex. NWT states for “not worse than” in Kreps notation. Equivalently: if $x \succeq z$ and $y \succeq z$, then $\alpha x + (1 - \alpha)y \succeq z$ for all $\alpha \in [0, 1]$.

Proposition (Characterization via $U(z)$). A preference relation $\succeq$ is convex $\iff$ each $U(z)$ is convex.

⁵ *Intuition for future me:* Under the product order, “more of each good never hurts.” If a representing u decreased in some component, it could invert $x \geq x'$ rankings. The statement uses that strictly increasing transforms preserve $\succeq$.

⁶ *Intuition for future me.* **GI:** for every bundle x there is *some* x' (possibly far) with $x' \succ x$. **LI:** for every x and every $\varepsilon > 0$ there is x' within ε of x with $x' \succ x$ —an improvement *arbitrarily close*.

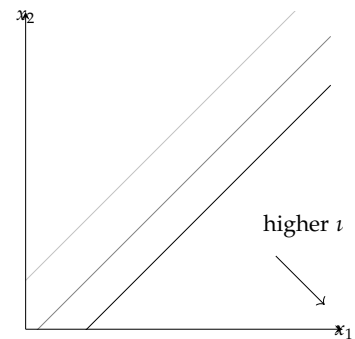


Figure 3: GI without monotonicity: $u = x_1 - x_2$.

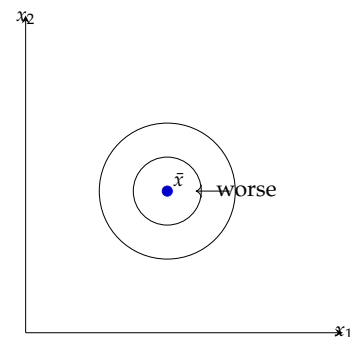


Figure 4: Local satiation (bliss point): LI fails at $\bar{x}$.

Proposition (Quasiconcave representation). *If $\succeq$ is convex and has a utility representation, then there exists a representing utility u that is quasiconcave (its upper contour sets $\{x : u(x) \geq t\}$ are convex).*

Intuition: Convexity encodes “love for averages”: if two bundles are at least as good as z , any mixture is too. That is exactly the convex-upper-set property of a quasiconcave u .

Concavity vs. convex preferences

Proposition (Concave utility $\Rightarrow$ convex preferences). *If a preference relation $\succeq$ is represented by a concave utility u , then the preference relation $\succeq$ is convex.*

One-line proof idea. For any z , the upper contour set $U(z) = \{x : u(x) \geq u(z)\}$ is convex because for $x, y \in U(z)$ and $\alpha \in [0, 1]$,

$$u(\alpha x + (1 - \alpha)y) \geq \alpha u(x) + (1 - \alpha)u(y) \geq u(z),$$

so $\alpha x + (1 - \alpha)y \in U(z)$. By Prop. , the preference relation $\succeq$ is convex.⁷ $\square$

Remark (The converse need not hold). Convex preferences admit a *quasiconcave* representation (Prop.), but *not every* representing utility is concave. E.g., if u is concave, then $v = \exp(u)$ is strictly increasing (represents the same order) but typically *not* concave. Quasiconcavity is the right ordinal notion.

Quasiconcavity and transformations

Definition (Quasiconcavity). A function $u : X \rightarrow \mathbb{R}$ is *quasiconcave* if for all $x, x' \in X$ and $\alpha \in [0, 1]$,

$$u(\alpha x + (1 - \alpha)x') \geq \min\{u(x), u(x')\}.$$

Equivalently, every *upper level set* $\{x : u(x) \geq c\}$ is convex.

Proposition (Convex preferences $\iff$ quasiconcave utility). *Suppose $\succeq$ has a utility representation. Then $\succeq$ is convex iff every utility representing $\succeq$ is quasiconcave.*

Why “every”? If u is quasiconcave and f is strictly increasing, then $v = f \circ u$ has the same upper sets up to a relabeling of thresholds:

$$\{x : v(x) \geq c\} = \{x : u(x) \geq f^{-1}(c)\},$$

so v is also quasiconcave. Since any two utilities that represent the same weak order differ by a strictly increasing transform, the property is invariant across *all* representatives. $\square$

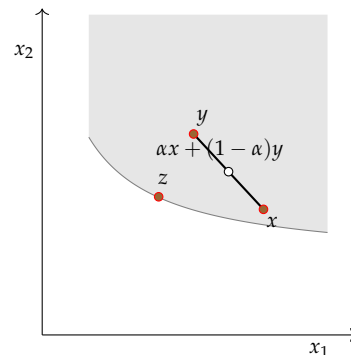


Figure 5: Convexity: the weakly preferred set $U(z)$ is convex. If $x, y \in U(z)$, then every mixture $\alpha x + (1 - \alpha)y$ also belongs to $U(z)$.

⁷ *Intuition for future me:* Concavity gives “diminishing marginal utility.” Super-level sets of a concave map are convex, hence mixtures are no worse than z —exactly convex preferences.

Remark (Handy facts). • If $X \subseteq \mathbb{R}$ and u is increasing, then u is quasi-concave. Indeed, for every c , the upper contour set

$$\{x \in X : u(x) \geq c\}$$

is an interval (and, if $X = \mathbb{R}_+$, a ray of the form $[a_c, \infty)$).

- Concavity implies quasiconcavity, but the converse is false.

Choice from convex sets under convex preferences

For any feasible set $A \subseteq X$, define the set of (weak) optima

$$C_{\succeq}(A) := \{x \in A : x \succeq x' \text{ for all } x' \in A\}.$$

Proposition ($C_{\succeq}(A)$ is convex). If A is convex and $\succeq$ is convex, then $C_{\succeq}(A)$ is convex.

Proof sketch. Take $x, y \in C_{\succeq}(A)$. Then for every $x' \in A$,

$$x \succeq x' \quad \text{and} \quad y \succeq x'.$$

By convexity of $\succeq$,

$$\alpha x + (1 - \alpha)y \succeq x' \quad \forall \alpha \in [0, 1].$$

Since A is convex, we also have $\alpha x + (1 - \alpha)y \in A$. Hence $\alpha x + (1 - \alpha)y$ is feasible and weakly preferred to every feasible bundle, so

$$\alpha x + (1 - \alpha)y \in C_{\succeq}(A).$$

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□

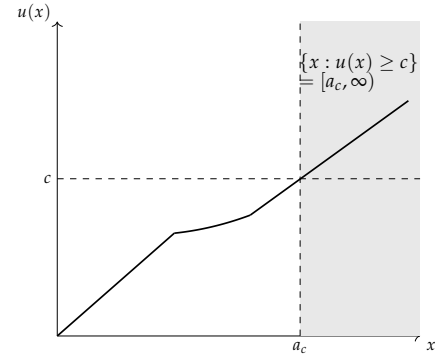


Figure 6: An increasing utility in one dimension is quasiconcave: for every level c , the upper contour set $\{x : u(x) \geq c\}$ is an interval (here, a ray $[a_c, \infty)$). The function drawn is increasing but not concave.

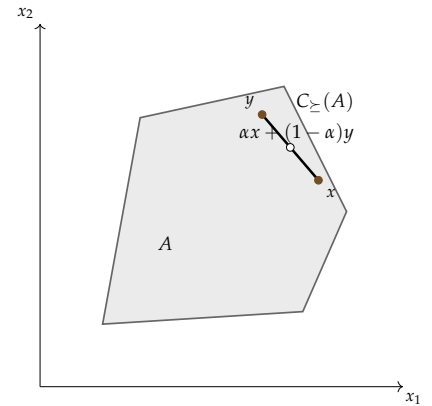


Figure 7: If A is convex and preferences are convex, then the set of optimal bundles $C_{\succeq}(A)$ is convex: whenever $x, y \in C_{\succeq}(A)$, every mixture $\alpha x + (1 - \alpha)y$ also belongs to $C_{\succeq}(A)$.

⁸ *Intuition for future me:* If two bundles beat every feasible rival, then any average of them also beats every feasible rival. Hence the whole set of optimal bundles is convex.

Appendix to Class 3: Recall of Formal Mathematical Concepts

Class 4: Regularity of Preferences

Roadmap

Preferences are the primitives. When they satisfy regularity properties, *some* (ordinally meaningful) properties can be imposed on a representing utility as well. We work on $X = \mathbb{R}_+^k$, with a complete and transitive $\succeq$.

Convexity (love for averages)

Definition (Convex preferences). A preference relation $\succeq$ is *convex* if for every $z \in X$, the weak upper contour set

$$U(z) := \{x \in X : x \succeq z\}$$

is convex. Equivalently: if $x \succeq z$ and $y \succeq z$, then $\alpha x + (1 - \alpha)y \succeq z$ for all $\alpha \in [0, 1]$.⁹

Semi-strict and strict convexity

There are three common strengths (chains hold left $\Rightarrow$ right):

$$\text{strictly convex} \Rightarrow \text{semi-strictly convex} \Rightarrow \text{convex}.$$

Definition (Semi-strict convexity). A preference relation $\succeq$ is *semi-strictly convex* if they are *convex* and for all x, y with $x \succeq y$ and $x \neq y$,

$$\alpha x + (1 - \alpha)y \succ y \quad \forall \alpha \in (0, 1).$$

Mixing something *weakly better* with y makes you *strictly* better than y .

Joaquín's comment: Semi-strict convexity

The condition in the definition should use $x \succ y$, not $x \succeq y$ together with $x \neq y$.

⁹ Intuition for future me: "Love for averages": if two bundles are no worse than z , any mixture is also no worse than z . Upper contour sets look convex (bulging toward the origin).

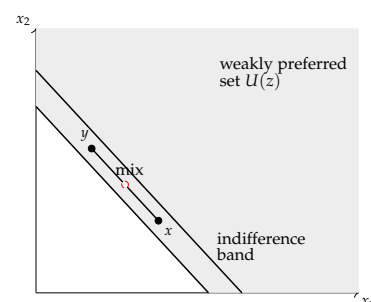


Figure 8: Convex but not semi-strict: the weakly preferred set $U(z)$ is convex, but the boundary contains a *fat indifference band*.

Indeed,

$$x \succeq y, x \neq y$$

includes both the cases $x \succ y$ and $x \sim y$ (with $x \neq y$). Therefore the condition

$$x \succeq y, x \neq y \implies \alpha x + (1 - \alpha)y \succ y$$

requires strict improvement even when x and y are distinct but indifferent. This is precisely the defining feature of *strict convexity*, not semi-strict convexity.

The correct definition is

$$x \succ y \implies \alpha x + (1 - \alpha)y \succ y, \quad \forall \alpha \in (0, 1),$$

together with convexity.

Intuitively, semi-strict convexity only strengthens convexity when one endpoint is already strictly preferred to the other. It rules out “thick” indifference sets: if $x \succ y$, then every interior mixture must be strictly better than y , so a region of indifference cannot exist. However, it still allows distinct indifferent bundles ($x \sim y$) to be connected by a flat indifference curve. Strict convexity goes one step further: even if $x \sim y$, every strict mixture must be strictly preferred, ruling out flat line segments on indifference curves and forcing them to be strictly curved.

Definition (Strict convexity). A preference relation $\succeq$ is *strictly convex* if for all $x \neq y$ with $x \succeq y$,

$$\alpha x + (1 - \alpha)y \succ y \quad \forall \alpha \in (0, 1).$$

In particular, if $x \sim y$ and $x \neq y$, every mixture strictly improves on y .

Remark (Semi-strict + continuity $\implies$ convex). If the preference relation $\succeq$ is *continuous* (next section), semi-strict convexity implies convexity. Intuition: the one-sided strict improvement along chords, plus closed upper sets, forces the whole upper set to be convex.

Proposition (Uniqueness of demand with strict convexity). If $A \subseteq X$ is convex (e.g., a budget set) and the preference relation $\succeq$ is strictly convex, then $C_{\succeq}(A)$ is a singleton. With LI or monotonicity and $p \gg 0$, the maximizer lies on $p \cdot x = y$.

Quasiconcavity (QC), semi-strict QC, and strict QC

Definition (Quasiconcavity (QC)). A function $u : X \rightarrow \mathbb{R}$ is *quasiconcave* if for all $x, y \in X$ and $\alpha \in [0, 1]$,

$$u(\alpha x + (1 - \alpha)y) \geq \min\{u(x), u(y)\}.$$

Equivalently, every upper contour set $\{x : u(x) \geq c\}$ is convex.

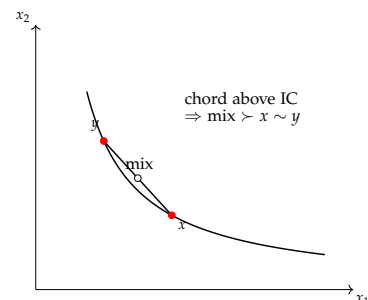


Figure 9: Strict convexity: if $x \sim y$ and $x \neq y$, then every interior mixture lies strictly above the indifference curve, and is therefore strictly preferred.

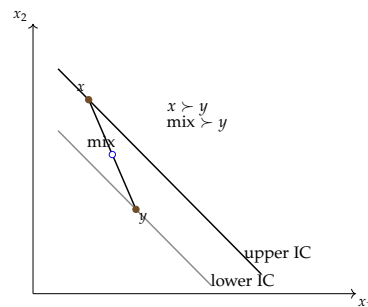


Figure 10: Semi-strictly convex but not strictly convex preferences. Indifference curves are linear (so strict convexity fails), but if $x \succ y$, then every interior mixture of x and y is still strictly preferred to y .

Definition (Semi-strict quasiconcavity). A function $u : X \rightarrow \mathbb{R}$ is *semi-strictly quasiconcave* if whenever $u(x) > u(y)$ and $\alpha \in (0, 1)$,

$$u(\alpha x + (1 - \alpha)y) > u(y).$$

Mixing a strictly better point with a worse one gives a strict improvement over the worse one.

Definition (Strict quasiconcavity). A function $u : X \rightarrow \mathbb{R}$ is *strictly quasiconcave* if for all $x \neq y$ and $\alpha \in (0, 1)$,

$$u(\alpha x + (1 - \alpha)y) > \min\{u(x), u(y)\}.$$

Equivalently: if $u(x) = u(y)$ with $x \neq y$, then any strict mixture has strictly *higher* utility.

Proposition (Invariance under monotone transforms). If u is (semi-strictly / strictly) quasiconcave and $f : \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing, then $f \circ u$ is also (semi-strictly / strictly) quasiconcave.

Proposition (Preferences vs. QC of utilities). Assume the preference relation $\succeq$ is represented by some u .

- (a) $\succeq$ is convex $\iff$ every representing utility is quasiconcave.
- (b) If u is strictly quasiconcave, then $\succeq$ is strictly convex.
- (c) If $\succeq$ is strictly convex and continuous, then there exists a representing utility that is strictly quasiconcave.

Proposition (Maximizers on convex sets). Let $A \subseteq X$ be convex.

- (a) If u is quasiconcave, the maximizer set $\arg \max_{x \in A} u(x)$ is convex.
- (b) If u is strictly quasiconcave, then $\arg \max_{x \in A} u(x)$ is a singleton.

Why concavity $\Rightarrow$ quasiconcavity (but not conversely)? Concavity imposes the stronger Jensen inequality:

$$u(\alpha x + (1 - \alpha)y) \geq \alpha u(x) + (1 - \alpha)u(y) \geq \min\{u(x), u(y)\},$$

so every concave function is quasiconcave. The converse fails because QC only requires the *weaker* “min” inequality (convex upper sets), which allows functions that curve *up* but have convex upper sets.

Relation to (semi-)strict convexity of preferences. Semi-strict convexity of $\succeq$ corresponds to semi-strict quasiconcavity of some representative: if $x \succ y$ and $x \neq y$, then mixing them strictly improves on y ; in utility terms, $u(\alpha x + (1 - \alpha)y) > u(y)$. Strict convexity of $\succeq$ corresponds to strict QC of (some) representing u , and—by invariance—of *every* strictly increasing transform of it.

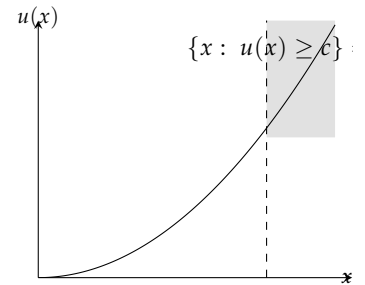


Figure 11: Quasiconcave but not concave: convex upper sets, graph curves upward.

Continuity

Definition (Continuity of preferences). A preference relation $\succeq$ is *continuous* if for all $x \succ y$ there exists $\varepsilon > 0$ such that whenever $\|x' - x\| \leq \varepsilon$ and $\|y' - y\| \leq \varepsilon$, we still have $x' \succ y'$. Equivalently (on $X \subseteq \mathbb{R}^k$): all upper and lower contour sets are closed.¹⁰

Joaquín's Comment: Sequential characterization of continuity

The neighborhood definition is the standard one, but the following equivalent sequential characterizations are often more convenient in proofs and, especially, when constructing counterexamples.

For complete and transitive preferences on $X = \mathbb{R}_+^L$, continuity is equivalent to:

1. If $x_n \rightarrow x$ and $y_n \rightarrow y$, then

$$x_n \succeq y_n \quad \forall n \implies x \succeq y.$$

2. Equivalently, fixing one bundle,

$$x_n \rightarrow x, x_n \succeq y \quad \forall n \implies x \succeq y,$$

and

$$x_n \rightarrow x, y \succeq x_n \quad \forall n \implies y \succeq x.$$

These formulations are equivalent to the neighborhood definition of continuity and are particularly useful when proving continuity or constructing counterexamples.

¹⁰ Intuition for future me: "Small perturbations don't flip strict rankings." No isolated jumps: ICs move smoothly (possibly with kinks).

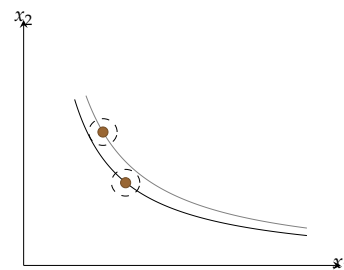


Figure 12: Continuity: small balls around $x \succ y$ keep the ranking.

Proposition (Continuous utility on $\mathbb{R}_+^k$ (Debreu)). On $X = \mathbb{R}_+^k$, if a preference relation $\succeq$ is continuous $\iff$ there exists a continuous utility u that represents $\succeq$.

Remark (What this does and does not say). If a continuous u represents $\succeq$, then $\succeq$ is continuous. If $\succeq$ is continuous, there exists a continuous representative u .

However, it is **not** true that *every* utility representing a continuous $\succeq$ must be continuous. Any strictly increasing (possibly discontinuous) transform of a representative also represents $\succeq$.

Homotheticity

Definition (Homothetic preferences). $\succeq$ is *homothetic* if for all $\lambda \geq 0$ and all x, y ,

$$x \succeq y \implies \lambda x \succeq \lambda y.$$

"Scaling all goods by the same factor preserves rankings."

Example. Cobb–Douglas $u(x) = x_1^\alpha x_2^\beta$ is homogeneous of degree $\alpha + \beta$ and represents homothetic preferences. Perfect substitutes $u = a^\top x$ and perfect complements $u = \min_i \{a_i x_i\}$ are homogeneous of degree 1 (also homothetic).

Proposition (Homogeneous representation). *If a preference relation $\succeq$ is continuous and homothetic on $\mathbb{R}_+^k$, then it admits a representation by a homogeneous utility u of some degree $\rho > 0$, i.e. $u(\lambda x) = \lambda^\rho u(x)$ for all $\lambda \geq 0$. (By a monotone transform, one can normalize to degree 1.)¹¹*

¹¹ Intuition for future me: Homothetic $\Rightarrow$ indifference sets are radial blow-ups of one another; MRS depends on *proportions*, not on scale.

Quasi-linear preferences (a money good)

Definition (Quasi-linearity in good k). $\succeq$ is *quasi-linear in x_k* if there exists $v : \mathbb{R}_+^{k-1} \rightarrow \mathbb{R}$ such that a representing utility can be written

$$u(x) = v(x_{-k}) + x_k, \quad x = (x_{-k}, x_k).$$

Proposition (Characterization). *On $\mathbb{R}_+^k$, the following are equivalent:¹²*

- (a) A preference relation $\succeq$ is continuous and quasi-linear in good k .
- (b) (i) Monotonicity in the numéraire (x_k): for fixed x_{-k} , $x'_k > x_k \Rightarrow (x_{-k}, x'_k) \succ (x_{-k}, x_k)$.
- (ii) Translation invariance in the numéraire: for all $c \in \mathbb{R}$,

$$(x_{-k}, x_k) \succeq (y_{-k}, y_k) \iff (x_{-k}, x_k + c) \succeq (y_{-k}, y_k + c).$$

- (c) Compensability via the numéraire: for every $x_{-k}, y_{-k} \in \mathbb{R}_+^{k-1}$ there exist $x_k, y_k \in \mathbb{R}_+$ such that

$$(x_{-k}, x_k) \sim (y_{-k}, y_k).$$

Intuition. Utility is additively separable and linear in the numéraire: money shifts utility one-for-one.

- (a) With x_{-k} fixed, more money is strictly better.
- (b) Adding the same c to money in both bundles preserves their ranking.
- (c) Differences in non-numéraire goods can be offset by transferring money so the bundles can be made indifferent.

Joaquín's comment: Correct characterization of quasi-linear preferences.

The proposition as stated is not correct. There are two distinct problems. First, conditions (b) and (c) are not separately equivalent to quasi-linearity. They must be imposed jointly. Second, even their conjunction does not

¹² Intuition for future me: Quasi-linearity means “money” enters additively: adding the same c units of the numéraire to both bundles never changes the ranking. Consequence in demand: no income effects for the non-numéraire goods.

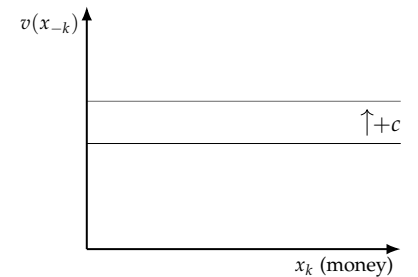


Figure 13: Quasi-linear: adding c to money shifts ICs vertically.

imply continuity of preferences. Continuity must either be assumed separately or included explicitly in the characterization.

Throughout, write a bundle as

$$(x, m) \in \mathbb{R}_+^{k-1} \times \mathbb{R}_+,$$

where m is the amount of the numéraire good.

Because the domain is $\mathbb{R}_+^k$, translation invariance should be understood only for translations that keep both bundles in the domain. Thus, for every $c \in \mathbb{R}$ such that

$$m + c \geq 0 \quad \text{and} \quad m' + c \geq 0,$$

one requires

$$(x, m) \succeq (y, m') \iff (x, m + c) \succeq (y, m' + c).$$

Why condition (b) alone is insufficient.

Consider lexicographic preferences on $\mathbb{R}_+ \times \mathbb{R}_+$, with the first good ranked before the numéraire:

$$(x, m) \succeq_{\text{lex}} (y, m') \iff [x > y] \text{ or } [x = y \text{ and } m \geq m'].$$

These preferences are strictly monotone in the numéraire. Moreover, they are translation invariant in the numéraire, because adding the same number to m and m' does not change either the comparison of x and y , or, when $x = y$, the comparison of m and m' .

But they cannot be quasi-linear given they do not admit a utility representation (see MWG).

Nevertheless, compensability fails. If $x > y$, then

$$(x, m) \succ_{\text{lex}} (y, m') \quad \text{for every } m, m' \in \mathbb{R}_+.$$

Hence no amount of the numéraire can compensate for the difference between x and y .

Why condition (c) alone is insufficient.

Consider Cobb–Douglas preferences on $\mathbb{R}_+^2$, represented by

$$U(x, m) = xm,$$

where m is the numéraire commodity.

Condition (c) holds. Indeed, let $x, y \in \mathbb{R}_+$. Choosing

$$m = y \quad \text{and} \quad m' = x,$$

we obtain

$$U(x, m) = xy = yx = U(y, m'),$$

so that

$$(x, y) \sim (y, x).$$

Hence, for every pair $x, y \in \mathbb{R}_+$, there exist $m, m' \in \mathbb{R}_+$ such that

$$(x, m) \sim (y, m').$$

Thus compensability, condition (c), is satisfied.

However, condition (b) fails.

First, strict monotonicity in the numéraire fails. At $x = 0$,

$$U(0, m) = 0 \quad \text{for every } m \in \mathbb{R}_+,$$

so whenever $m > m'$,

$$(0, m) \sim (0, m'),$$

rather than

$$(0, m) \succ (0, m').$$

Second, translation invariance fails. Since

$$U(1, 2) = 2 = U(2, 1),$$

we have

$$(1, 2) \sim (2, 1).$$

After adding one unit of the numéraire to both bundles,

$$U(1, 3) = 3 \quad \text{whereas} \quad U(2, 2) = 4,$$

so

$$(2, 2) \succ (1, 3).$$

Hence

$$(1, 2) \sim (2, 1), \quad \text{but} \quad (1, 3) \not\sim (2, 2),$$

contradicting translation invariance.

Finally, although Cobb–Douglas preferences are continuous, they are not quasi-linear, since they fail condition (b). Therefore, condition (c) alone implies neither condition (b) nor condition (a) (continuous and quasi-linear preferences).

Why (b) and (c) together do not imply continuity.

Consider preferences on $\mathbb{R}_+^2$ represented by

$$U(x, m) = \ln(1 + x) + \mathbf{1}\{x > 1\} + m,$$

where m is the numéraire commodity. Equivalently,

$$U(x, m) = \begin{cases} \ln(1 + x) + m, & 0 \leq x \leq 1, \\ \ln(1 + x) + 1 + m, & x > 1. \end{cases}$$

These preferences satisfy condition (b). First, they are strictly monotone in the numéraire: if $m' > m$, then

$$U(x, m') - U(x, m) = m' - m > 0,$$

and therefore

$$(x, m') \succ (x, m).$$

Second, they satisfy translation invariance in the numéraire. For every $c \in \mathbb{R}$ such that the translated bundles remain in $\mathbb{R}_+^2$,

$$\begin{aligned} (x, m) \succeq (y, m') &\iff U(x, m) \geq U(y, m') \\ &\iff U(x, m) + c \geq U(y, m') + c \\ &\iff U(x, m + c) \geq U(y, m' + c) \\ &\iff (x, m + c) \succeq (y, m' + c). \end{aligned}$$

Condition (c) also holds. Define

$$v(x) := \ln(1+x) + \mathbf{1}\{x > 1\}.$$

For arbitrary $x, y \in \mathbb{R}_+$, choose

$$m = \max\{v(y) - v(x), 0\}, \quad m' = \max\{v(x) - v(y), 0\}.$$

Then $m, m' \geq 0$ and

$$v(x) + m = v(y) + m',$$

so

$$(x, m) \sim (y, m').$$

Hence the preferences satisfy compensability.

Nevertheless, the preference relation is not continuous. To prove this directly, let

$$z_n = \left(1 + \frac{1}{n}, 0\right), \quad z = (1, 0), \quad y = \left(1, \frac{1}{2}\right).$$

For every n ,

$$U(z_n) = \ln\left(2 + \frac{1}{n}\right) + 1 > \ln 2 + \frac{1}{2} = U(y),$$

and therefore

$$z_n \succ y.$$

Moreover,

$$z_n \longrightarrow z.$$

At the limit, however, the indicator term is equal to zero, so

$$U(z) = \ln 2 < \ln 2 + \frac{1}{2} = U(y),$$

and hence

$$y \succ z.$$

Thus the upper contour set of y ,

$$\mathcal{U}(y) := \left\{w \in \mathbb{R}_+^2 : w \succeq y\right\},$$

contains every element of the sequence $\{z_n\}$, but does not contain its limit z . Therefore $\mathcal{U}(y)$ is not closed, and the preference relation is not continuous. Hence conditions (b) and (c) characterize quasi-linearity, but they do not imply continuity.

Proposition (Characterization of quasi-linear preferences). *Let $\succeq$ be a complete and transitive preference relation on*

$$X = \mathbb{R}_+^{k-1} \times \mathbb{R}_+.$$

The following are equivalent:

- (i) *The preferences admit a quasi-linear representation in the numéraire: there exists a function*

$$v : \mathbb{R}_+^{k-1} \rightarrow \mathbb{R}$$

such that

$$(x, m) \succeq (y, m') \iff v(x) + m \geq v(y) + m'.$$

(ii) The following three properties hold jointly:

(a) Ordering by the numéraire: for every $x \in \mathbb{R}_+^{k-1}$ and every $m, m' \in \mathbb{R}_+$,

$$(x, m) \succeq (x, m') \iff m \geq m'.$$

(b) Translation invariance in the numéraire: for every $x, y \in \mathbb{R}_+^{k-1}$, every $m, m' \in \mathbb{R}_+$, and every $c \in \mathbb{R}_+$,

$$(x, m) \succeq (y, m') \iff (x, m + c) \succeq (y, m' + c).$$

(c) Compensability through the numéraire: for every $x, y \in \mathbb{R}_+^{k-1}$, there exist $m, m' \in \mathbb{R}_+$ such that

$$(x, m) \sim (y, m').$$

Moreover, $\succeq$ is continuous and quasi-linear if and only if it is continuous and satisfies (ii)(a)–(c).

Proof. We prove the two implications separately.

(i) implies (ii).

Suppose that

$$U(x, m) = v(x) + m$$

represents $\succeq$.

Strict monotonicity in the numéraire is immediate. If $m' > m$, then

$$U(x, m') - U(x, m) = m' - m > 0,$$

and therefore

$$(x, m') \succ (x, m).$$

For translation invariance, let c be such that both translated bundles remain in the domain. Then

$$\begin{aligned} (x, m) \succeq (y, m') &\iff v(x) + m \geq v(y) + m' \\ &\iff v(x) + m + c \geq v(y) + m' + c \\ &\iff (x, m + c) \succeq (y, m' + c). \end{aligned}$$

Finally, compensability follows because utility differences in the non-numéraire goods can always be offset by the numéraire. Given x and y , define

$$m = \max\{v(y) - v(x), 0\}, \quad m' = \max\{v(x) - v(y), 0\}.$$

Then $m, m' \geq 0$ and

$$v(x) + m = v(y) + m',$$

so

$$(x, m) \sim (y, m').$$

(ii) implies (i).

Fix an arbitrary reference bundle

$$\bar{x} \in \mathbb{R}_+^{k-1}.$$

By compensability, for every $x \in \mathbb{R}_+^{k-1}$, there exist $a_x, b_x \in \mathbb{R}_+$ such that

$$(x, a_x) \sim (\bar{x}, b_x).$$

Define

$$v(x) := b_x - a_x.$$

We first show that $v(x)$ is well defined, namely, that it does not depend on the particular compensating pair chosen. Suppose also that

$$(x, \tilde{a}_x) \sim (\bar{x}, \tilde{b}_x).$$

Using translation invariance, add $\tilde{a}_x$ to the first indifference and add a_x to the second. This gives

$$(x, a_x + \tilde{a}_x) \sim (\bar{x}, b_x + \tilde{a}_x)$$

and

$$(x, a_x + \tilde{a}_x) \sim (\bar{x}, \tilde{b}_x + a_x).$$

By transitivity of indifference,

$$(\bar{x}, b_x + \tilde{a}_x) \sim (\bar{x}, \tilde{b}_x + a_x).$$

Strict monotonicity in the numéraire implies that two bundles with the same non-numéraire component can be indifferent only if their numéraire components are equal. Hence

$$b_x + \tilde{a}_x = \tilde{b}_x + a_x,$$

or equivalently,

$$b_x - a_x = \tilde{b}_x - \tilde{a}_x.$$

Thus v is well defined.

Now take arbitrary bundles (x, m) and (y, m') . Choose $L \geq 0$ large enough that

$$m + L \geq a_x \quad \text{and} \quad m' + L \geq a_y.$$

Translation invariance applied to

$$(x, a_x) \sim (\bar{x}, b_x)$$

with translation $m + L - a_x \geq 0$ yields

$$(x, m + L) \sim (\bar{x}, b_x + m + L - a_x) = (\bar{x}, v(x) + m + L).$$

Similarly,

$$(y, m' + L) \sim (\bar{x}, v(y) + m' + L).$$

Therefore, by transitivity,

$$\begin{aligned} (x, m + L) \succeq (y, m' + L) &\iff (\bar{x}, v(x) + m + L) \succeq (\bar{x}, v(y) + m' + L) \\ &\iff v(x) + m + L \geq v(y) + m' + L \\ &\iff v(x) + m \geq v(y) + m'. \end{aligned}$$

The second equivalence follows from condition (ii)(a), since bundles with the same non-numéraire component are ranked exactly according to their amounts of the numéraire. Finally, translation invariance gives

$$(x, m) \succeq (y, m') \iff (x, m + L) \succeq (y, m' + L).$$

Combining the preceding equivalences,

$$(x, m) \succeq (y, m') \iff v(x) + m \geq v(y) + m'.$$

Thus $U(x, m) = v(x) + m$ represents $\succeq$, proving quasi-linearity.

The final statement follows immediately: the three properties characterize quasi-linearity, while continuity is an additional, logically independent requirement. $\square$

Appendix to Class 4: Curvature of u and induced preference convexity.

Property of u	Upper contour sets	Preferences induced	Argmax on convex sets (demand)
Strictly concave	Strictly convex	Strictly convex	Unique maximizer
Semi-strictly concave	Convex; strictness only when $u(x) \neq u(y)$	Convex (not necessarily strict)	Possibly multiple
Concave	Convex	Convex	Possibly multiple
Strictly quasiconcave	Strictly convex	Strictly convex	Unique maximizer
Semi-strictly quasiconcave	Convex; strict on chords across indifference	Strictly convex	Unique maximizer
Quasiconcave	Convex	Convex	Possibly multiple

Key facts. Concavity $\Rightarrow$ QC. Strict versions (strictly concave / strictly QC / semi-strictly QC) $\Rightarrow$ strictly convex preferences $\Rightarrow$ uniqueness of the maximizer on convex feasible sets (e.g., budget sets). Non-strict versions yield convex (not strictly) preferences and allow multiplicity.

Outside the six cases (what if u is none of them?)

- **Continuous but not QC:** upper contour sets need not be convex $\Rightarrow$ preferences are continuous but may be non-convex; maximizer sets on convex budgets can be disconnected or large; uniqueness not guaranteed.
- **Strictly increasing (monotone) but not continuous/QC:** preferences are monotone (more is better) but may fail continuity and convexity; existence/continuity of demand can fail without additional assumptions (e.g., closed/compact feasibility).
- **Arbitrary representatives (discontinuous transforms):** preserve ordinal rankings but can destroy regularity properties of the representative (continuity, differentiability) without changing the underlying (possibly regular) preferences.

Operational takeaway. For uniqueness and clean comparative statics, target **strict convexity of preferences** (achieved if u is strictly concave, strictly QC, or semi-strictly QC). With only convexity (concave / semi-strictly concave / QC), expect potential multiplicity.

Class 5: Separability and Choice vs. Demand

Additive separability (two blocks)

Separability asks when preferences can be decomposed into “blocks” of goods so that trade-offs within a block do not depend on the quantities of goods outside the block. In the simplest two-block *additively separable* case, overall utility is the sum of a subutility for each block; cross-block substitution disappears, and the interaction across blocks operates only through the budget.

$$u(x_1, \dots, x_\ell, x_{\ell+1}, \dots, x_k) = v(x_1, \dots, x_\ell) + \omega(x_{\ell+1}, \dots, x_k).$$

Definition (Weak separability into blocks $Y_1, \dots, Y_N$). Let $Y_n \subseteq \{1, \dots, k\}$ be nonempty and pairwise disjoint ($Y_n \cap Y_{\tilde{n}} = \emptyset$ for $n \neq \tilde{n}$). Write $x_{Y_n} \in \mathbb{R}_+^{|Y_n|}$ for the subvector of x on coordinates in Y_n , and Y_n^c for the complement in $\{1, \dots, k\}$.

A preference relation $\succeq$ on $\mathbb{R}_+^k$ is *weakly separable into* $Y_1, \dots, Y_N$ if there exist subutility functions $V_n : \mathbb{R}_+^{|Y_n|} \rightarrow \mathbb{R}$ and an aggregator f that is increasing in each V_n such that

$$u(x) = f(V_1(x_{Y_1}), V_2(x_{Y_2}), \dots, V_N(x_{Y_N}), x_{(Y_1 \cup \dots \cup Y_N)^c}),$$

and, equivalently, for every n , all $x_{Y_n}, \hat{x}_{Y_n} \in \mathbb{R}_+^{|Y_n|}$, and all common complements $x_{Y_n^c}, \hat{x}_{Y_n^c} \in \mathbb{R}_+^{|Y_n^c|}$ with $x_{Y_n^c} = \hat{x}_{Y_n^c}$,

$$(x_{Y_n}, x_{Y_n^c}) \succeq (\hat{x}_{Y_n}, x_{Y_n^c}) \iff V_n(x_{Y_n}) \geq V_n(\hat{x}_{Y_n}).$$

Acá hay que ver bien que se dice y poner intuición.

Definition (Strong separability (additive across blocks)). Let $Y_1, \dots, Y_N$ form a partition of $\{1, \dots, k\}$ (i.e., pairwise disjoint and $\bigcup_{n=1}^N Y_n = \{1, \dots, k\}$). A preference relation $\succeq$ is *strongly separable into* $Y_1, \dots, Y_N$ if it is weakly separable into this partition and admits the additive representation

$$u(x) = V_1(x_{Y_1}) + V_2(x_{Y_2}) + \dots + V_N(x_{Y_N}).$$

Joaquín's comment: Strong separability vs. additive separability

The definition above actually describes *additive separability*, not *strong separability*. Following Kreps (and adapting it to Tilman's notation), these are different concepts: strong separability is a property of the *preference relation*, whereas additive separability is a property of a *utility representation*.

Let

$$I = \{1, \dots, n\}$$

denote the commodity index set, and let

$$\mathcal{P} = \{P \subseteq I : P \neq \emptyset\}$$

be the collection of all nonempty subsets of goods.

A subset $P \in \mathcal{P}$ is said to be *weakly separable* if, for every pair of bundles x_P, x'_P and every pair of outside bundles x_{-P}, x'_{-P} ,

$$(x_P, x_{-P}) \succeq (x'_P, x_{-P}) \implies (x_P, x'_{-P}) \succeq (x'_P, x'_{-P}).$$

That is, the ranking of bundles that differ only in the goods of P is independent of the consumption of goods outside P .

Define

$$\hat{\mathcal{P}} = \{P \in \mathcal{P} : P \text{ is weakly separable}\}.$$

Thus, $\hat{\mathcal{P}}$ is the collection of all weakly separable subsets of goods.

A collection

$$\hat{\mathcal{P}}^* \subseteq \hat{\mathcal{P}}$$

is called a *strongly separable partition* (equivalently, preferences $\succeq$ are **strongly separable in $\hat{\mathcal{P}}^*$**) if:

- (i) $\hat{\mathcal{P}}^*$ is a partition of the commodity index set (its elements are pairwise disjoint and their union is the entire set of goods); and
- (ii) $\hat{\mathcal{P}}^*$ is *closed under unions*: whenever $A, B \in \hat{\mathcal{P}}^*$, then

$$A \cup B \in \hat{\mathcal{P}}^*.$$

Notice that by definition, $\hat{\mathcal{P}}^* \subseteq \hat{\mathcal{P}}$, so every block of the partition is weakly separable. Closure under unions therefore means that not only each individual block is weakly separable, but every union of blocks is also weakly separable. Thus, if

$$L = A \cup B \cup \dots, \quad A, B, \dots \in \hat{\mathcal{P}}^*,$$

then $L \in \hat{\mathcal{P}}$, so for every such union,

$$(x_L, x_{L^c}) \succeq (x'_L, x_{L^c})$$

for some outside bundle x_{L^c} implies

$$(x_L, x'_{L^c}) \succeq (x'_L, x'_{L^c})$$

for every outside bundle x'_{L^c} . In other words, the ranking of bundles within L is independent of the consumption of goods outside L .

Only afterwards does Kreps prove that, under additional assumptions (continuity and nontriviality on at least three partition blocks), strongly separable preferences admit an additive representation,

$$u(x) = \sum_{p \in \mathcal{P}^*} u_p(x_p).$$

Therefore, additive separability should be presented as a representation theorem, not as the definition of strong separability.

Choice vs. Demand

Let $X \subseteq \mathbb{R}_+^k$ be the consumption set and let $\mathcal{E}$ denote the collection of all nonempty subsets of X .

Definition (Choice correspondence). A *choice correspondence* is a map $C : \mathcal{E} \rightrightarrows X$ with $C(B) \subseteq B$ for every $B \in \mathcal{E}$.

Definition (Budget set). For $p \in \mathbb{R}_{++}^k$ and $y > 0$,

$$B(p, y) = \{x \in \mathbb{R}_+^k : p \cdot x \leq y\}.$$

Definition (Demand correspondence / function). Given a preference relation $\succeq$ on X , the *demand correspondence* is

$$D(p, y) = \arg \max_{x \in B(p, y)} u(x) \quad \text{for any utility } u \text{ representing } \succeq.$$

When $D(p, y)$ is a singleton, we write $x(p, y)$ for its unique element and call it the *demand function*.

Consumer problem and basic properties

Definition (Consumer problem (CP)). Given $p \in \mathbb{R}_{++}^k$ and $y > 0$,

$$\max_{x \in \mathbb{R}_+^k} u(x) \quad \text{s.t. } p \cdot x \leq y,$$

and $D(p, y)$ denotes the set of solutions.

Proposition. Suppose the preference relation $\succeq$ is continuous (equivalently, admits a continuous utility u).

- (i) For every $p \in \mathbb{R}_{++}^k$ and $y > 0$, the CP has at least one solution: $D(p, y) \neq \emptyset$.
- (ii) If $x \in D(p, y)$ and $\lambda > 0$, then $x \in D(\lambda p, \lambda y)$.
- (iii) If u is quasi-concave (i.e., $\succeq$ is convex), then $D(p, y)$ is a convex set. If u is strictly quasi-concave, then $D(p, y)$ is a singleton.

(iv) If $\succeq$ is locally non-satiated, then every $x \in D(p, y)$ satisfies the budget:
 $p \cdot x = y$.

Sketch for (iii). Let $x, y \in D(p, y)$ and $\lambda \in [0, 1]$. *Feasibility:* the budget set is convex, so

$$p \cdot (\lambda x + (1 - \lambda)y) = \lambda p \cdot x + (1 - \lambda) p \cdot y \leq y,$$

hence $z := \lambda x + (1 - \lambda)y \in B(p, y)$.

Optimality under QC: by quasi-concavity,

$$u(z) \geq \min\{u(x), u(y)\}.$$

Since x, y are optimal, they both attain the same maximal utility level:

$$u(x) = u(y) = M := \max_{a \in B(p, y)} u(a).$$

By quasi-concavity, then,

$$u(z) \geq M.$$

But z is feasible, so it cannot yield utility strictly above the maximum M . Hence $u(z) = M$, and therefore $z \in D(p, y)$.

Uniqueness under strict QC: if $x \neq y$, strict quasi-concavity gives

$$u(\lambda x + (1 - \lambda)y) > \min\{u(x), u(y)\} = \max_{B(p, y)} u,$$

a contradiction. Hence $D(p, y)$ is a *singleton*. □

Appendix to class 5: Summary & Intuition

Weak separability (by blocks). Preferences are weakly separable into a partition $\{Y_1, \dots, Y_N\}$ when the trade-offs *within* a block do not depend on quantities *outside* the block. Equivalently, there exist subutilities $V_n(x_{Y_n})$ and an aggregator f (increasing in each V_n) such that $u(x) = f(V_1, \dots, V_N, x_{(Y_1 \cup \dots \cup Y_N)^c})$.

Operational test (MRS-independence): for $i, j \in Y_n$,

$$\frac{\partial}{\partial x_k} \left(\frac{u_i}{u_j} \right) = 0 \quad \forall k \notin Y_n,$$

so the within-block MRS is unaffected by goods outside Y_n .

Intuition: choice can be done in stages—collapse each block to an index, then compare blocks via the budget.

Strong separability. Strong (additive) separability strengthens weak separability to

$$u(x) = \sum_{n=1}^N V_n(x_{Y_n}).$$

Quick checks: (i) If $u \in C^2$ and all cross-block second derivatives vanish, $\partial^2 u / \partial x_i \partial x_j = 0$ for $i \in Y_r, j \in Y_s, r \neq s$, then u is additively separable across blocks. (ii) If $u = f(V_1, \dots, V_N)$ with a CES or product aggregator, a monotone transform (e.g. $\phi = \log$ or a power) makes $\phi \circ u$ additively separable across blocks.

Choice vs. demand (quick facts). Let $B(p, y) = \{x \geq 0 : p \cdot x \leq y\}$ and $D(p, y) = \arg \max_{x \in B(p, y)} u(x)$. Under continuity: existence $D(p, y) \neq \emptyset$. Scaling: if $x \in D(p, y)$ then $x \in D(\lambda p, \lambda y)$ (homogeneity of degree zero). Under (strict) quasi-concavity: $D(p, y)$ is convex (a singleton if strict). With local non-satiation: any $x \in D(p, y)$ exhausts the budget $p \cdot x = y$.

What to write in proofs. (1) State the partition and apply the MRS-independence test for weak separability. (2) For strong separability, either verify vanishing cross-block Hessian terms (additivity) or exhibit a suitable monotone transform that turns the index aggregator into a sum.

Class 6: Consumer Problem (CP)

Setup.

- Consumers are price takers.
- Choice: $x \in \mathbb{R}_+^k$.
- Standard conditions yield a well-behaved optimization problem.
- Sets we use below are compact (closed and bounded).

Definition (Budget set). For $p \in \mathbb{R}_{++}^k$ and $y > 0$,

$$B(p, y) = \{x \in \mathbb{R}_+^k : p \cdot x \leq y\}.$$

Definition (Demand correspondence and indirect utility). For any $p \in \mathbb{R}_{++}^k$ and $y > 0$, let

$$D(p, y) = \arg \max_{x \in B(p, y)} u(x) \quad \text{and} \quad v(p, y) = u(x^*) \quad \text{for any } x^* \in D(p, y).$$

$D(p, y)$ is a (possibly set-valued) demand correspondence; $v(p, y)$ is the indirect utility.

Proposition (Basic properties of the CP). *Given a compact budget set, $B(p, y)$,*

- (i) *If u is continuous, then at least one solution exists: $D(p, y) \neq \emptyset$.¹³*
- (ii) *If u is quasi-concave, then the set of solutions $D(p, y)$ is convex.*
- (iii) *If u is strictly quasi-concave, then there is either no solution or exactly one solution.*
- (iv) *If preferences are locally non-satiated, then any solution x^* satisfies $p \cdot x^* = y$.*
- (v) *If $x^* \in D(p, y)$ and $\lambda > 0$, then $x^* \in D(\lambda p, \lambda y)$.*

¹³ Look that this is simply Weierstrass Theorem.

Proposition (Homogeneity of degree zero). *For all $\lambda > 0$,*

$$D(\lambda p, \lambda y) = D(p, y) \quad \text{and} \quad v(\lambda p, \lambda y) = v(p, y).$$

Demand and indirect utility

For any $p \in \mathbb{R}_{++}^k$ and $y > 0$, let $D(p, y)$ be the set of all solutions to the consumer problem. If $D(p, y) \neq \emptyset$, define the *indirect utility*

$$v(p, y) = u(x^*) \quad \text{for any } x^* \in D(p, y).$$

Note that $D(p, y)$ is a (possibly set-valued) *demand correspondence* and $v(p, y)$ is the *indirect utility function*.

Proposition (Homogeneity of degree zero). *For every $\lambda > 0$,*

$$D(\lambda p, \lambda y) = D(p, y) \quad \text{and} \quad v(\lambda p, \lambda y) = v(p, y).$$

Intuition: rescaling both prices and income by the same factor leaves the set of solutions unchanged.

Proposition. *If u is continuous, then:*

- (i) D is nonempty-valued;
- (ii) D is upper hemicontinuous (the relevant continuity notion for correspondences);
- (iii) v is continuous.

Theorem 2 (Berge's Maximum Theorem). *If $f : T \times X \rightarrow \mathbb{R}$ is continuous and $\Gamma : T \rightrightarrows X$ is nonempty, compact-valued and continuous (upper and lower hemicontinuous), then the value function $v(t) = \max_{x \in \Gamma(t)} f(t, x)$ is continuous and the argmax $A(t) = \arg \max_{x \in \Gamma(t)} f(t, x)$ is nonempty, compact-valued, and upper hemicontinuous.*

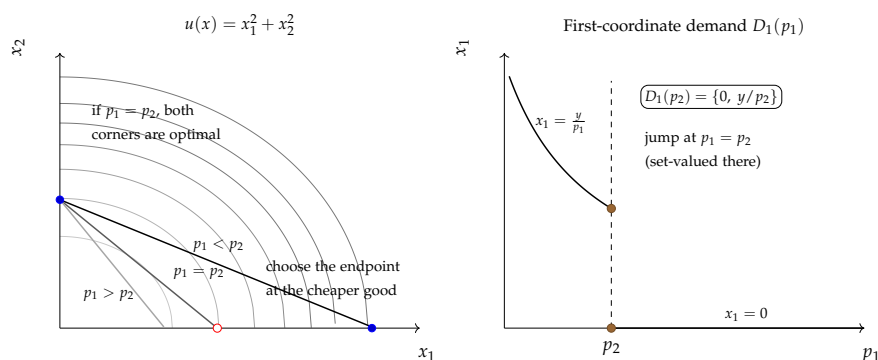


Figure 14: Example with non-quasi-concave utility $u(x) = x_1^2 + x_2^2$. Left: the consumer chooses the corner corresponding to the cheaper good; when $p_1 = p_2$, both corners are optimal. Right: the first-coordinate demand correspondence $D_1(p_1)$ jumps at $p_1 = p_2$ and is set-valued there.

Upper hemicontinuity: jumps versus discontinuities

Definition (Upper hemicontinuity (UHC)). A correspondence $F : \Theta \rightrightarrows X$ is *upper hemicontinuous at θ^** if for every sequence $\theta_n \rightarrow \theta^*$ and every

sequence $x_n \in F(\theta_n)$ such that $x_n \rightarrow x^*$, we have $x^* \in F(\theta^*)$. We say F is UHC if this holds at every θ^* .

Definition (UHC of Marshallian demand). Let $L \in \mathbb{N}$, $\Theta = \mathbb{R}_{++}^L \times \mathbb{R}_+$ with elements (p, y) , and $X = \mathbb{R}_+^L$. The (Marshallian) demand correspondence is $D : \Theta \rightrightarrows X$. We say that D is *upper hemicontinuous at (p^*, y^*)* if for every sequence $(p^n, y^n) \rightarrow (p^*, y^*)$ and every sequence $x^n \in D(p^n, y^n)$ such that $x^n \rightarrow x^*$, we have $x^* \in D(p^*, y^*)$. We say D is UHC if this holds for every $(p^*, y^*) \in \Theta$.

Remark (Intuition). As prices and income (p, y) vary slightly, track any optimal bundles $x_n \in D(p_n, y_n)$. If $x_n \rightarrow x^*$, upper hemicontinuity says x^* remains optimal at the limit (p^*, y^*) . **Heuristic:** the set of optimal choices can *shrink* in the limit but cannot *sprout new points outside*; limit points of optimal selections never “jump out” of the limiting demand set.

Remark (How Kreps phrases it). In Euclidean spaces Kreps uses the sequential version above (often called *upper semicontinuity*). Intuitively: limit points of selections cannot “jump outside” the limiting set; values may shrink but not expand.

Proposition (Closed-graph characterization). *With the sequential notion of UHC used here, a correspondence F is UHC if and only if its graph $\text{Gr}(F) = \{(\theta, x) : x \in F(\theta)\}$ is sequentially closed. In finite-dimensional Euclidean spaces, this is equivalent to the graph being closed.*

Proposition (Demand is UHC under continuity). *If u is continuous, then for every $y > 0$ the demand correspondence $D(\cdot, y) : \mathbb{R}_{++}^k \rightrightarrows \mathbb{R}_+^k$ is nonempty, compact-valued, and UHC. (Proof idea: Berge’s Maximum Theorem.)*

Remark (Intuition). Upper hemicontinuity is weaker than continuity. It does not say that the demand set varies smoothly with prices. It only says that if a sequence of optimal bundles converges, then its limit remains optimal at the limiting prices. Hence UHC is compatible with jumps and with set-valued demand at knife-edge prices.

Intuition.

- UHC as “no escaping limits”: if $p^n \rightarrow p^*$ and $x^n \in D(p^n, y)$ with $x^n \rightarrow x^*$, then the limit bundle must still be optimal:

$$x^* \in D(p^*, y).$$

Thus, limits of optimal selections cannot fall outside the limiting demand set.

- *What UHC allows:* the demand set at the limit may be larger than nearby demand sets. This is exactly what happens at a tie: when $p_1 = p_2$, several bundles may suddenly become optimal.
- *What UHC rules out:* it rules out an “outward jump” in which optimal bundles along nearby prices converge to a bundle that is not optimal at the limiting price.
- *Closed graph:* in the sequential formulation used here, UHC says precisely that whenever $(p^n, x^n) \rightarrow (p^*, x^*)$ with $x^n \in D(p^n, y)$, then (p^*, x^*) also belongs to the graph of D .

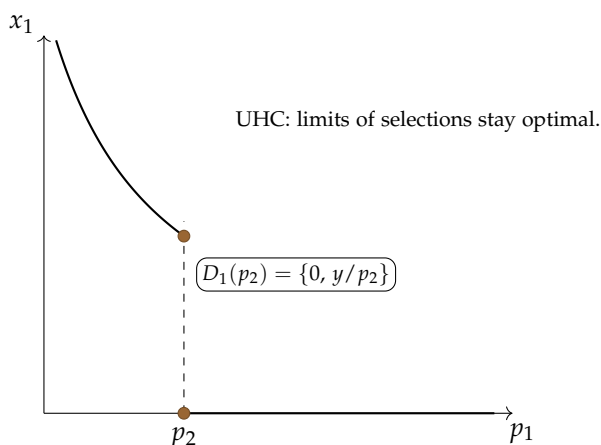


Figure 15: UHC in the non-quasi-concave jump case. When $p_1 < p_2$, the consumer demands only good 1; when $p_1 > p_2$, she demands none of good 1; and at $p_1 = p_2$ both corner demands are optimal. The graph has a jump, but every convergent sequence of optimal selections has a limit inside $D(p^*, y)$.

Remark (UHC versus convex-valued demand). The two figures illustrate different issues. In the non-quasi-concave example, demand is UHC but not convex-valued at the tie: $D_1(p_2) = \{0, y/p_2\}$. In the perfect-substitutes case, demand is UHC and convex-valued at the tie: $D_1(p_2) = [0, y/p_2]$. Thus quasi-concavity does not eliminate set-valued demand, but it makes the demand correspondence convex-valued.

Remark (Why UHC matters for equilibria). Fixed-point theorems used for existence typically require correspondences that are nonempty, compact-valued and UHC (e.g., Kakutani also needs convex-valued). If demand were not UHC, limit arguments in equilibrium proofs can fail; even with UHC alone, additional properties (e.g., convexity of values, aggregate feasibility) are needed to close the existence proof.

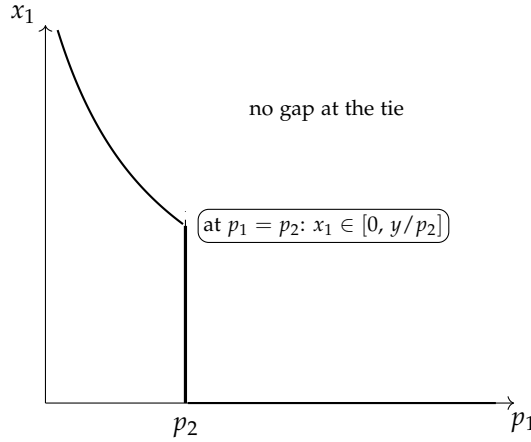


Figure 16: Perfect substitutes. At $p_1 = p_2$, the whole budget line is optimal, so the first-coordinate demand is the interval $[0, y/p_2]$. The vertical segment is filled in: demand is convex-valued at the tie, unlike in the non-quasi-concave example.

Differentiable case: KKT/FOC

Assume $u : \mathbb{R}_+^k \rightarrow \mathbb{R}$ is differentiable. Consider

$$\max_{x \in \mathbb{R}_+^k} u(x) \quad \text{s.t.} \quad p \cdot x \leq y, \quad x \geq 0.$$

Lagrangian:

$$\mathcal{L}(x, \lambda, \mu) = u(x) + \lambda(y - p \cdot x) + \mu \cdot x, \quad \lambda \geq 0, \mu \geq 0.$$

Definition (First-order (KKT) conditions). A feasible x^* satisfies the FOC/KKT if there exist multipliers $\lambda^* \geq 0$ and $\mu^* \geq 0$ such that

$$\begin{aligned} \text{(stationarity)} \quad & \nabla u(x^*) - \lambda^* p - \mu^* = 0, \\ \text{(primal feas.)} \quad & p \cdot x^* \leq y, \quad x^* \geq 0, \\ \text{(dual feas.)} \quad & \lambda^* \geq 0, \quad \mu^* \geq 0, \\ \text{(compl. slackness)} \quad & \lambda^* (y - p \cdot x^*) = 0, \quad \mu_j^* x_j^* = 0 \quad \forall j. \end{aligned}$$

Equivalently, for each j ,

$$\frac{\partial u}{\partial x_j}(x^*) \leq \lambda^* p_j, \quad \text{with equality if } x_j^* > 0, \quad \text{and} \quad p \cdot x^* \leq y, \quad \text{with equality if } \lambda^* > 0.$$

Joaquín's comment: Consistent KKT sign convention

The stationarity condition is inconsistent with the Lagrangian. To use nonnegative KKT multipliers, write the nonnegativity constraints as

$$-x \leq 0,$$

rather than $x \geq 0$. Then the stated lagrangian is correct,

$$\mathcal{L}(x, \lambda, \mu) = u(x) - \lambda(p \cdot x - y) - \mu \cdot (-x) = u(x) + \lambda(y - p \cdot x) + \mu \cdot x,$$

but the correct **stationarity condition** is

$$\nabla u(x^*) - \lambda^* p + \mu^* = 0.$$

The remaining KKT conditions are unchanged.

Proposition. *Two important points:*

- (i) If x^* solves the consumer problem, then x^* satisfies the FOC/KKT.
- (ii) If u is concave and the constraints are convex, then any x^* that satisfies the FOC/KKT is (globally) optimal. In particular, if there is a unique KKT solution, it is the unique global maximizer.

At the optimum (marginal utility per dollar). From the KKT,

$$\frac{1}{p_i} \frac{\partial u}{\partial x_i}(x^*) \leq \lambda^* \quad \text{for all } i, \quad \text{and if } x_i^* > 0: \quad \frac{1}{p_i} \frac{\partial u}{\partial x_i}(x^*) = \lambda^*.$$

Hence, for any goods i, j that are consumed in positive amounts,

$$\frac{1}{p_i} \frac{\partial u}{\partial x_i}(x^*) = \frac{1}{p_j} \frac{\partial u}{\partial x_j}(x^*) = \lambda^* \quad (\text{"marginal utility in dollars" equalized}).$$

Envelope (income). If the indirect utility $v(p, y)$ is differentiable in y , then by the envelope theorem

$$\frac{\partial v(p, y)}{\partial y} = \lambda^*,$$

where λ^* is the multiplier at the optimum corresponding to (p, y) .

Example: $u(x_1, x_2) = x_1 + \sqrt{x_2}$

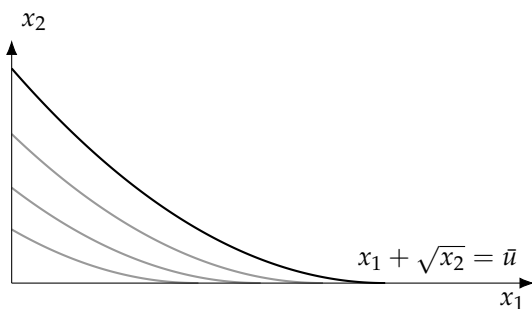


Figure 17: Indifference curve for $u(x_1, x_2) = x_1 + \sqrt{x_2}$ at level $\bar{u}$: $x_2 = (\bar{u} - x_1)^2$.

With prices $p = (p_1, p_2) \in \mathbb{R}_{++}^2$ and income $y > 0$, let $\lambda \geq 0$ be the multiplier on $p \cdot x \leq y$ and $\mu \geq 0$ on $x \geq 0$. The KKT conditions are, for $j = 1, 2$,

$$\frac{\partial u}{\partial x_j}(x^*) \leq \lambda p_j \quad (= \text{if } x_j^* > 0), \quad p \cdot x^* \leq y \quad (= \text{if } \lambda > 0).$$

Here

$$\frac{\partial u}{\partial x_1} = 1, \quad \frac{\partial u}{\partial x_2} = \frac{1}{2\sqrt{x_2}} \quad (x_2 > 0).$$

Case 1 (interior: $x_1^* > 0, x_2^* > 0$). Equalities hold:

$$1 = \lambda p_1 \text{ (which implies } \lambda > 0) \quad \frac{1}{2\sqrt{x_2^*}} = \lambda p_2 \Rightarrow \lambda = \frac{1}{p_1}, \quad x_2^* = \left(\frac{p_1}{2p_2}\right)^2.$$

Binding budget gives

$$x_1^* = \frac{y}{p_1} - \frac{p_1}{4p_2}.$$

Feasibility requires $x_1^* \geq 0$, i.e.

$$y \geq \frac{p_1^2}{4p_2}.$$

Case 2 (corner: $x_1^* = 0, x_2^* > 0$). Then $p_2 x_2^* = y \Rightarrow x_2^* = \frac{y}{p_2}$ and

$$1 \leq \lambda p_1, \quad \frac{1}{2\sqrt{x_2^*}} = \lambda p_2 \Rightarrow \lambda = \frac{1}{2\sqrt{p_2 y}}, \quad 1 \leq \frac{p_1}{2\sqrt{p_2 y}} \Leftrightarrow y \leq \frac{p_1^2}{4p_2}.$$

Demand correspondence.

$$D(p_1, p_2, y) = \begin{cases} \left(\frac{y}{p_1} - \frac{p_1}{4p_2}, \left(\frac{p_1}{2p_2} \right)^2 \right), & \text{if } y \geq \frac{p_1^2}{4p_2}, \\ \left(0, \frac{y}{p_2} \right), & \text{if } y \leq \frac{p_1^2}{4p_2}. \end{cases}$$

(At $y = \frac{p_1^2}{4p_2}$ both expressions coincide.)

“MU per dollar” equalization (interior). If $x_1^*, x_2^* > 0$ then

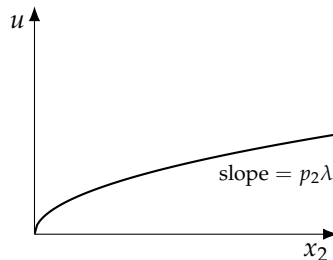
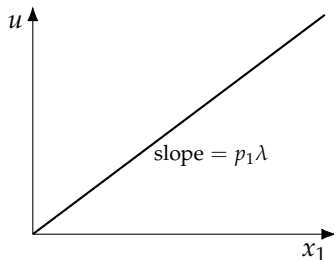
$$\frac{1}{p_1} = \frac{1}{p_2} \cdot \frac{1}{2\sqrt{x_2^*}} \Rightarrow x_2^* = \left(\frac{p_1}{2p_2} \right)^2.$$

Whether the consumer buys good 1 depends on prices and income. In particular, the consumer consumes only good 2 whenever

$$y \leq \frac{p_1^2}{4p_2} \Rightarrow p_1 \geq 2\sqrt{yp_2}.$$

In this case, good 1 is sufficiently expensive relative to income and the price of good 2, so the consumer spends all income on good 2.

Conditional on being interior, the optimal quantity of good 2 depends only on prices, not on income. Once the consumer reaches this target level of x_2 , additional income is spent entirely on good 1.



Appendix to Class 6: Solving the Consumer Problem — A General Playbook

Standing notation. Prices $p \in \mathbb{R}_{++}^k$, income $y > 0$, choice $x \in \mathbb{R}_+^k$,

- Budget set $B(p, y) = \{x \geq 0 : p \cdot x \leq y\}$
- Demand correspondence $D(p, y) \subseteq \arg \max_{x \in B(p, y)} u(x)$
- Indirect utility $v(p, y) = \max_{x \in B(p, y)} u(x)$.

What graders expect to see (principles before math)

- (a) **Meta-properties first.** State continuity / (strict) quasi-concavity of u and convexity/compactness of $B(p, y)$. This pins down existence, convexity of D , and (under strict qc) uniqueness, i.e. $D(p, y)$ is unique-valued (a marshallian demand function). If u is concave and constraints are convex, KKT are *necessary and sufficient*.

- If u is continuous and $B(p, y)$ is nonempty and compact, then by **Weierstrass** a maximizer exists: $D(p, y) \neq \emptyset$.
- If u is quasi-concave, then $D(p, y)$ is convex (possibly set-valued); if u is strictly quasi-concave, any maximizer is unique.
- If u is concave (or strictly concave) and constraints are convex, then any KKT point is a global maximizer (KKT are necessary and sufficient; uniqueness if strict concavity holds).
- If u is not quasi-concave (e.g., convex utility), KKT are *only necessary*: generate interior and corner candidates via KKT and then compare $u(\cdot)$ across candidates.

KKT quick implications by utility class (with optimality status).

- **Cobb–Douglas / CES (concave).** Interior solution via MU/ p equalization; then use $p \cdot x = y$ to get closed-form shares. *KKT status: necessary & sufficient* (concave objective, convex feasible set). *Uniqueness:* yes if strictly qc (e.g., all weights > 0 ; for CES in the concave range). *Corners:* only if parameters force a good to drop out (weight = 0) or feasibility binds at nonnegativity.
- **Quasi-linear (x_1 as numeraire).** $\lambda^* = 1/p_1$; solve the non-linear block with MU/ $p = 1/p_1$ and allocate residual income to x_1 . *KKT status: necessary & sufficient* if $v(\cdot)$ is concave. *Uniqueness:* yes if the non-linear block is strictly concave. *Corners:* arise when income is too small to hit the interior for the non-linear block.
- **Perfect substitutes ($u = \sum a_i x_i$).** Compare a_i/p_i ; spend all income on any argmax. At ties, $D(p, y)$ is the full budget segment spanned

Why KKT are necessary & sufficient (intuition).

In a concave maximization over a convex feasible set, the “surface” of u is bowl-shaped downward: any local improvement is also a global one. KKT encode two ideas: (i) *no profitable local move* (the gradient cannot point into the feasible set at the optimum, except along flat directions), and (ii) *which constraints pinch* (complementary slackness identifies binding constraints and zeros). Equalizing MU per dollar means a one-dollar reallocation cannot raise u ; with concavity, “no local gain” $\Rightarrow$ “no global gain”. Conversely, any true maximizer must satisfy the same trade-offs and pinching, hence KKT are also necessary.

Necessary vs. sufficient (plain words).

Necessary: every true optimum must satisfy KKT; if a point violates KKT, it cannot be optimal. *Sufficient:* any point that satisfies KKT is (globally) optimal. *Necessary & sufficient:* equivalence (KKT is both a filter and a certificate). If only necessary (non-concave u), KKT can produce *false positives* (e.g., saddles), so you **must compare** utilities across KKT-generated candidates.

Slater (CQ) for the Consumer Problem.

General: For convex programs $\max u(x)$ s.t. $g_i(x) \leq 0$, Slater holds if there exists a point with all $g_i(x) < 0$ (strict feasibility). *Here:* $g_0(x) = p \cdot x - y \leq 0$ and $g_j(x) = -x_j \leq 0$. If $p \in \mathbb{R}_{++}^k$ and $y > 0$, pick $x = \varepsilon \mathbf{1} \gg 0$ with ε small: then $p \cdot x < y$ and $-x_j < 0$ for all $j \Rightarrow$ Slater holds. *Implications:* With concave u and linear constraints, strong duality; KKT are *necessary & sufficient* (use subgradients if u has kinks). *Relation to MFQC:* MFQC is a smooth first-order CQ. In convex problems, Slater $\Rightarrow$ MFQC and is easier to verify here. *Edge cases:* If $y = 0$ or some $p_i = 0$, strict feasibility may fail; handle separately (often a corner like $x = 0$).

by tied goods. *KKT status*: **necessary & sufficient** (linear $\equiv$ concave); may yield *many* KKT solutions at ties. *Uniqueness*: no at price-ratio ties (set-valued demand along the budget line). *Practical tie-handling*: report the convex set of budget-feasible mixtures of tied goods.

- **Perfect complements (Leontief, $u = \min\{x_i/\alpha_i\}$)**. Fixed proportions at the kink; the budget pins the scalar multiple. *KKT status*: with *subgradients*, **necessary & sufficient** (concave as a minimum of linear forms). *Uniqueness*: yes (unique kink bundle scaled by y). *Corners*: non-issue—solution sits at the kink (not an “interior” in the smooth sense).
- **Convex utilities (non-qc; e.g., $u = x_1^2 + x_2^2$)**. Interior FOCs do not characterize a maximum; solutions push to corners at the cheapest good(s). *KKT status*: **only necessary** (can flag stationary points that are *not* global maxima). *Workflow*: generate interior/corner candidates via KKT, then *compare* u and partition (p, y) by the verification inequalities. *Ties*: when price ratios equalize, demand is set-valued (entire budget segment).

Constraint qualification (Slater). In the consumer problem with $p \in \mathbb{R}_{++}^k$, $y > 0$, there exists $x \gg 0$ with $p \cdot x < y$; hence Slater holds. For concave u and convex constraints, KKT is *necessary and sufficient*. With kinks (nondifferentiable u), replace gradients by *subgradients* and the same logic applies.

- (b) **Budget binds under monotonicity**. If preferences are locally non-satiated (e.g., monotone in each good), then any optimizer satisfies $p \cdot x^* = y$.
- (c) **KKT written in “GSI format”**. For each j ,

$$\frac{\partial u}{\partial x_j}(x^*) \leq \lambda p_j \quad (= \text{if } x_j^* > 0), \quad p \cdot x^* \leq y \quad (= \text{if } \lambda > 0).$$

This line is your workhorse; it drives all the case analysis.

- (d) **Case-by-case structure**. Interior: equalize “MU per dollar” and use the binding budget to solve. Corners: set $x_k = 0$, solve for λ and remaining x , then *verify* the inequality on the zeroed good(s).
- (e) **Partition the parameter space**. Convert the verification inequalities into clean thresholds in (p, y) . Report $D(p, y)$ (and $v(p, y)$ if asked) *piece-wise*.
- (f) **If u is not concave**. Say it explicitly. Treat KKT as *necessary* only, enumerate candidates (interior/corners), and compare $u(\cdot)$ across candidates.
- (g) **One-line regularity checks**. Homogeneity of degree zero, Walras’ law ($p \cdot x(p, y) = y$ when monotone), monotonicity in y , and (when relevant) UHC via Berge.

Cookie-cutter for Marshallian demand (exam version)

1. *State properties*: u continuous (and, if true, concave/strictly qc); $B(p, y)$ convex/-compact. Therefore existence (and uniqueness if strict qc) of D .
2. *Write KKT succinctly*:

$$\nabla u(x^*) - \lambda p - \mu = 0, \quad p \cdot x^* \leq y, \quad x^* \geq 0$$

Equivalently: $\partial u / \partial x_j \leq \lambda p_j$ (= if $x_j^* > 0$) and budget binds if $\lambda > 0$.
3. *Show budget binds*: argue local non-satiation/monotonicity $\Rightarrow \lambda > 0$ and $p \cdot x^* = y$.
4. *Interior candidate*: if plausible, equalize MU per dollar $\frac{1}{p_i} \frac{\partial u}{\partial x_i} = \frac{1}{p_j} \frac{\partial u}{\partial x_j}$ for all i, j with $x_i^* > 0$, substitute into $p \cdot x = y$, solve x^* , check $x^* \geq 0$.
5. *Corner candidates*: choose subset S with $x_j = 0$ for $j \notin S$, solve on S with equalities, then verify $\partial u / \partial x_j \leq \lambda p_j$ at $j \notin S$. This yields threshold conditions in (p, y) .
6. *Partition and report*: translate verifications into clean regimes and give

$$D(p, y) = \begin{cases} x^{(1)}(p, y), & \text{if condition 1} \\ x^{(2)}(p, y), & \text{if condition 2} \\ \{x^{(2)}(p, y), x^{(3)}(p, y)\}, & \text{if ties} \end{cases}$$

7. Find $v(p, y) = u(x^{(\cdot)}(p, y))$.
8. (If asked) *regularity quick-checks*: homogeneity o, Walras' law, monotonicity in y , UHC via Berge.

Exam-time diagnostics (fast checks)

- **Homothetic** $u \Rightarrow$ Engel curves are straight lines through the origin; $x(p, y) = y h(p)$ where $h(p)$ is a vector of optimal consumption when income is 1, $h(p) := x(p, 1)$.
- **If a good is strictly essential** (e.g., complements): interior in that good unless y too small to reach the kink.
- **Strictly convex** u (non-qc) $\Rightarrow$ *corners* at the cheapest good; report ties as a set at price equalities.
- **MU/ p equalization** is the quickest route for interiors; the binding budget closes the system.
- **Tie-handling**: if two regimes meet at a threshold, list *both* optimal bundles at the threshold (set-valued demand).

Playbooks for typical $u(\cdot)$

(i) **Cobb–Douglas**. $u(x) = \prod_{i=1}^k x_i^{\alpha_i}$, $\alpha_i > 0$.

Interior (always, with $x_i^* > 0$): $\frac{\partial u / \partial x_i}{p_i} = \frac{\partial u / \partial x_j}{p_j} \Rightarrow x_i^* = \frac{\alpha_i}{\sum_{\ell} \alpha_{\ell}} \frac{y}{p_i}$.

Properties:

- x_i is homogeneous of degree 0 in (p, y) ;
- expenditure shares = $\alpha_i / \sum \alpha_\ell$

(ii) CES with elasticity $\sigma > 0, \sigma \neq 1$. Let $u(x) = (\sum_{i=1}^k a_i x_i^{\frac{\sigma-1}{\sigma}})^{\frac{\sigma}{\sigma-1}}$ with $a_i > 0$. Then the Marshallian demand is

$$x_i(p, y) = y \frac{a_i p_i^{-\sigma}}{\sum_{\ell=1}^k a_\ell p_\ell^{1-\sigma}}, \quad (\text{and as } \sigma \rightarrow 1 \text{ this converges to Cobb-Douglas}).$$

Equivalent share form: $s_i = \frac{p_i x_i}{y} = a_i \left(\frac{p_i}{P} \right)^{1-\sigma}$ with $P = (\sum_\ell a_\ell p_\ell^{1-\sigma})^{\frac{1}{1-\sigma}}$.

(iii) Quasi-linear (numeraire x_1). $u(x) = x_1 + v(x_2, \dots, x_k)$ with v concave, increasing.

$$\text{Rule: } \frac{\partial v}{\partial x_j}(x^*) = \lambda p_j \quad (j \geq 2), \quad \lambda = \frac{1}{p_1}.$$

Compute $(x_2^*, \dots, x_k^*)$ from $MU/p = 1/p_1$ and the budget for the non-linears; residual income goes to x_1 :

$$x_1^* = \frac{y}{p_1} - \sum_{j \geq 2} \frac{p_j}{p_1} x_j^* \geq 0.$$

If y is too small to reach the interior for the non-linear block, you hit a corner there and the remainder goes to the numeraire.

(iv) Perfect substitutes. $u(x) = \sum_i a_i x_i$ with $a_i > 0$. Pick any good with highest a_i / p_i . If a unique maximizer i^* exists,

$$x_{i^*}^* = \frac{y}{p_{i^*}}, \quad x_j^* = 0 \quad (j \neq i^*).$$

If ties at the top, $D(p, y)$ is the convex set of all bundles mixing tied goods on the budget line.

(v) Perfect complements (Leontief). $u(x) = \min\{x_1/\alpha_1, \dots, x_k/\alpha_k\}$ with $\alpha_i > 0$. Fixed proportions at the kink:

$$x^* = \theta (\alpha_1, \dots, \alpha_k), \quad \theta = \frac{y}{p \cdot \alpha} \Rightarrow x_i^* = \alpha_i \frac{y}{\sum_\ell p_\ell \alpha_\ell}.$$

(vi) Convex utilities (e.g., $u = x_1^2 + x_2^2$). Preferences are not quasi-concave; KKT are only necessary. Candidates are corners at the cheapest good:

$$x^* = \begin{cases} (y/p_1, 0), & p_1 < p_2, \\ (0, y/p_2), & p_2 < p_1, \\ \{(t, y/p_2 - (p_1/p_2)t) : t \in [0, y/p_1]\}, & p_1 = p_2. \end{cases}$$

Report demand piece-wise and be explicit about the set at $p_1 = p_2$.

(vii) “Power” or isoelastic blocks. For $u(x_1, x_2) = x_1^\beta + g(x_2)$ with $\beta \in (0, 1)$, the interior condition is $\beta x_1^{\beta-1}/p_1 = g'(x_2)/p_2$ with budget binding. Proceed by solving one variable from MU/p and substituting into $p \cdot x = y$.

Berge’s Maximum Theorem (consumer-specialized): existence, continuity, and demand regularity

Setup. Parameters $t := (p, y) \in \mathbb{R}_{++}^k \times \mathbb{R}_{++}$, feasible set $B(p, y) = \{x \in \mathbb{R}_+^k : p \cdot x \leq y\}$, objective $u : \mathbb{R}_+^k \rightarrow \mathbb{R}$ continuous. Define the demand correspondence $D(p, y) \subseteq \arg \max_{x \in B(p, y)} u(x)$ and indirect utility $v(p, y) := \max_{x \in B(p, y)} u(x)$.

Statement. If

- (i) u is continuous,
- (ii) for every (p, y) the feasible set $B(p, y)$ is nonempty and compact, and
- (iii) the correspondence $(p, y) \mapsto B(p, y)$ is continuous (both upper and lower hemicontinuous),

Then:

- (a) **Existence & compact argmax:** $D(p, y)$ is nonempty and compact-valued.
- (b) **Continuity of indirect utility:** $v(p, y)$ is continuous in (p, y) .
- (c) **Upper hemicontinuity (UHC) of demand:** $D(\cdot)$ is UHC in (p, y) .
- (d) **Convexity & uniqueness under curvature:** If u is (strictly) quasi-concave, $D(p, y)$ has convex (respectively, singleton) values; if u is strictly concave on a convex $B(p, y)$, D is a *continuous function*.

Compactness at a glance. If $p \gg 0$ and $y > 0$, then $0 \leq x_i \leq y/p_i$ for each i , hence $B(p, y)$ is a nonempty, closed, bounded (thus compact) polytope.

Why (iii) holds here. With linear constraints and $p \gg 0$, small changes in (p, y) deform $B(p, y)$ continuously; no facets “appear/disappear” discontinuously. This delivers the continuity of the feasible-set mapping needed by Berge.

Class 7: Comparative statistics and GARP (Chapter 4)

Setup and notation

Let $D(p, y) \subseteq \mathbb{R}_+^L$ be the *demand correspondence* of a decision maker, represented by some preferences (or utility). For each price vector $p \in \mathbb{R}_+^L$ and income $y \in \mathbb{R}_+$, the set $D(p, y)$ contains the *optimal consumption bundles*. We write $x \in D(p, y)$ to indicate an optimal choice at (p, y) .

¹⁴

¹⁴ *Intuition for future me:* Think of $D(\cdot)$ as “what the consumer buys” at (p, y) . We now ask what happens to demand when y , a good’s own price p_i , or a cross-price p_j changes. At this stage we do not predict any specific relation, we just classify possibilities.

What can happen when y , p_i , or p_j changes

- **If income y changes:** good i can be
 - *Normal*: as y increases (with prices fixed), demand for i increases.
 - *Inferior*: as y increases (with prices fixed), demand for i decreases.
- **If a cross price p_j changes (with $j \neq i$):** good i can be
 - *Substitute*: as p_j increases, x_i increases (and decreases if p_j falls).
 - *Complement*: as p_j increases, x_i decreases (and increases if p_j falls).
- **If the own price p_i changes:** demand for i can be
 - *Normal*: the usual downward-sloping demand.
 - *Giffen*: demand for i increases when p_i increases.

Towards characterization

In the end, we want to characterize changes in p_i or p_j with two different effects. For example, we will record a change in prices and then describe the set of options that are weakly preferred.

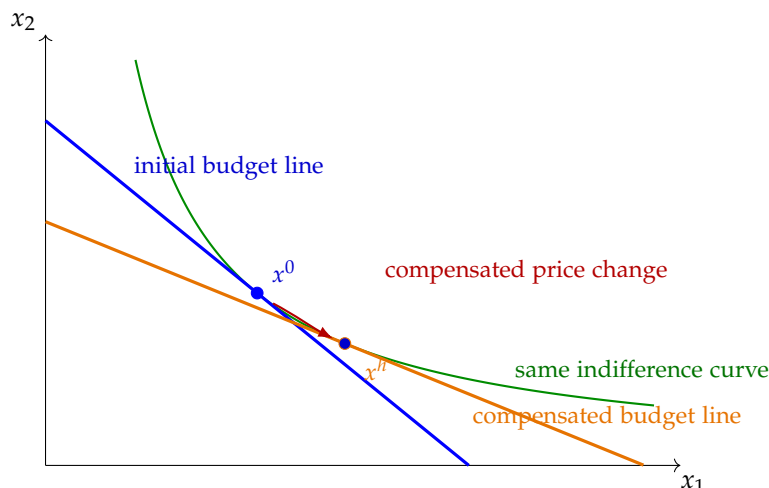


Figure 18: Compensated price change: the compensated budget line has the new relative prices but is tangent to the same indifference curve. Thus utility is held constant while the optimal bundle moves from x^0 to x^h .

Compensated Law of Demand and a consequence for Giffen goods

Remark. (Budget exhaustion under LNS). If preferences are locally non-satiated (LNS) and $x \in D(p, y)$, then the budget is exhausted: $p \cdot x = y$. Same for any $x' \in D(p', y')$.

Proposition (Compensated law of demand (Slutsky inequality)). Suppose preferences are LNS. Let $x \in D(p, y)$. Fix any $p' \in \mathbb{R}_+^L$ and define the Slutsky compensation $y' := p' \cdot x$. Let $x' \in D(p', y')$. Then

$$(p' - p) \cdot (x' - x) \leq 0.$$

Proof. By budget exhaustion and the compensation $y' = p' \cdot x$, we have

$$p' \cdot x' = y' = p' \cdot x \Rightarrow p' \cdot (x' - x) = 0.$$

Since $x' \in D(p', y')$ and x is affordable at (p', y') , we have $x' \succeq x$. Because x solves the (p, y) -problem and $x' \succeq x$, local non-satiation implies $p \cdot x' \geq p \cdot x$. Hence

$$(p' - p) \cdot (x' - x) = p' \cdot (x' - x) - p \cdot (x' - x) = 0 - (p \cdot x' - p \cdot x) \leq 0. \quad \square$$

Remark. This result is about *compensated* price changes. It does not contradict the possibility of Giffen behavior, which is an *uncompensated* (Marshallian) phenomenon driven by the income effect.

Proposition (Giffen $\Rightarrow$ Inferior). Suppose preferences are LNS. Fix (p, y) and let $x \in D(p, y)$. Consider a price change that raises only the own price of good i : $p'_i > p_i$ and $p'_j = p_j$ for $j \neq i$. Assume (Giffen behavior) that for the uncompensated problem at income y there exists $x^u \in D(p', y)$ with $x^u_i > x_i$. Define the compensated income $\bar{y} := p' \cdot x$ and pick $\bar{x} \in D(p', \bar{y})$.

Compensated monotonicity. This proposition is exactly that compensated (Hickian) demand is *decreasing* in prices. Hold utility fixed—stay on the same indifference curve—and ask which bundle is chosen. As prices tilt, the same-utility choice tilts the other way: less of what became relatively more expensive, more of what became relatively cheaper. This is a directional (not coordinatewise) monotonicity statement.

Scalar vs. coordinatewise. When we write $p \cdot x' \geq p \cdot x$, this is a *single-number* (scalar) cost comparison at prices p —not a coordinatewise comparison of x' and x . Likewise, $(p' - p) \cdot (x' - x) \leq 0$ is an *inner product*: it aggregates price and quantity changes into one number that captures whether quantities move, on net, *against* the price change.

Then

$$\bar{y} > y, \quad \bar{x}_i \leq x_i < x_i^u,$$

so at prices p' a higher income $\bar{y}$ yields (weakly) less of good i than a lower income y . Hence good i is (weakly) inferior.

Proof. Since only p_i increases and $x_i \geq 0$, we have $\bar{y} = p' \cdot x > p \cdot x = y$. By the compensated law of demand applied to (p, y) and $(p', \bar{y})$ we get $(p' - p) \cdot (\bar{x} - x) \leq 0$, which here reduces to

$$(p'_i - p_i)(\bar{x}_i - x_i) \leq 0 \Rightarrow \bar{x}_i \leq x_i.$$

By the Giffen assumption for the *uncompensated* choice at (p', y) , we have $x_i^u > x_i$. Putting these together,

$$\bar{x}_i \leq x_i < x_i^u,$$

so when income increases from y to $\bar{y}$ (holding p' fixed), demand for good i decreases. Thus i is inferior. $\square$

Data set

We observe a finite set of choices:

$$\{(x_1, p_1, y_1), (x_2, p_2, y_2), \dots, (x_n, p_n, y_n)\},$$

where each $x_i \in \mathbb{R}_+^L$ is the chosen bundle at prices $p_i \in \mathbb{R}_+^L$ and income $y_i = p_i \cdot x_i$ (by local non-satiation).

Revealed preference

Lemma. Let a preference relation $\succeq$ be complete, transitive, and locally non-satiated preferences. If $x^* \in D(p, y)$ is chosen at prices p with income y , then:

1. For every x with $p \cdot x = y$, we have $x^* \succeq x$.
2. For every x with $p \cdot x < y$, we have $x^* \succ x$.

Proof. (i) If $p \cdot x = y$, then x is feasible at (p, y) . Since x^* is chosen, $x^* \succeq x$. (ii) If $p \cdot x < y$, local non-satiation implies there exists x' arbitrarily close to x with $x' \succ x$ and $p \cdot x' \leq y$. Then x' is feasible, so $x^* \succeq x'$. By transitivity, $x^* \succ x$. $\square$

Definition (Direct revealed preference). Given an observation (x_i, p_i, y_i) with $y_i \geq p_i \cdot x_i$:

- *Weak direct revealed preference:* if $p_i \cdot x_j \leq y_i$, then $x_i \succeq^d x_j$.
- *Strict direct revealed preference:* if $p_i \cdot x_j < y_i$, then $x_i \succ^d x_j$.

Definition (Revealed preference (transitive closure)). We write $x_i \succeq^R x_j$ if there exist indices $k_1, \dots, k_m$ (possibly $m = 0$) such that

$$x_i \succeq^d x_{k_1}, \quad x_{k_1} \succeq^d x_{k_2}, \quad \dots, \quad x_{k_m} \succeq^d x_j.$$

If at least one step in the chain is strict, then $x_i \succ^R x_j$. In particular, allowing $m = 0$ gives $x_i \succeq^d x_j \Rightarrow x_i \succeq^R x_j$.

GARP

Definition (Generalized Axiom of Revealed Preference (GARP)). A data set satisfies GARP if there is no bundle x_i such that

$$x_i \succ^R x_i.$$

In words: revealed preference must not generate cycles with a strict step.

Afriat's Theorem

Theorem 3 (Afriat). *The following statements are equivalent:*

1. *The data set can be rationalized by a complete, transitive, locally non-satiated preference relation.*
2. *The data set satisfies GARP.*
3. *The data set can be rationalized by a complete, transitive, locally non-satiated, convex, continuous, monotone preference relation.*

GARP intuition. Think “no improvement loops.” (i) If a chosen bundle spends strictly less than income ($p^i \cdot x^i < y^i$), then it is *strictly* directly revealed preferred to itself ($x^i \succ^d x^i$), so GARP is immediately violated. (ii) More generally, if there is a chain of revealed weak preferences from x^i to x^j and another back from x^j to x^i , with at least one strict link, the two chains form a strict cycle ($x^i \succ^R x^i$). GARP = no such cycles.

Class 8 – Choice under Uncertainty (Kreps Ch. 5)

We now move from deterministic choice to situations under uncertainty. The key idea is to represent actions by probability distributions over outcomes, and to assume that agents maximize expected utility.

Outcomes and Probability Distributions

Let X denote the set of outcomes. For simplicity, assume X is finite.

Definition (Probability Distributions over Outcomes). We denote by $\Delta(X)$ the set of all probability distributions over X . Each $p \in \Delta(X)$ is a function $p : X \rightarrow [0, 1]$ with $\sum_{x \in X} p(x) = 1$.

Definition (Degenerate Distribution). For each $x \in X$, let $\delta_x \in \Delta(X)$ be the degenerate (Dirac) distribution that assigns probability 1 to outcome x and 0 to all other outcomes.¹⁵

¹⁵ *Intuition for future me:* We are identifying actions with distributions over outcomes. Deterministic choices are special cases represented by δ_x .

Preferences over Lotteries

Definition (Preference relation over lotteries). Let X be the set of outcomes and let $\Delta(X)$ be the set of lotteries over X . A preference relation over lotteries is a binary relation

$$\succsim \subseteq \Delta(X) \times \Delta(X).$$

Thus, for lotteries $p, q \in \Delta(X)$, we write $p \succsim q$ whenever $(p, q) \in \succsim$, meaning that lottery p is weakly preferred to lottery q .

Properties of the Preference Relation

We impose axioms on $\succsim$ analogous to the deterministic setting:

- (i) **Completeness and Transitivity.** The relation $\succsim$ is complete and transitive.¹⁶
- (ii) **Continuity.** Preferences are continuous: if $p \succ q \succ r$, then small perturbations of p or r preserve the ranking.

¹⁶ *Intuition for future me:* The agent can rank all lotteries consistently.

Under these two axioms, we already know that preferences can be represented by a utility function $U : \Delta(X) \rightarrow \mathbb{R}$ such that

$$p \succsim q \iff U(p) \geq U(q).$$

The Independence Axiom

We add a third crucial axiom:

Definition (Independence). For all lotteries $p, q, r \in \Delta(X)$ and all $\alpha \in (0, 1)$,

$$p \succsim q \implies \alpha p + (1 - \alpha)r \succsim \alpha q + (1 - \alpha)r.$$

This axiom captures the idea that preferences respect probabilistic mixtures.¹⁷

Representation by Expected Utility

Theorem 4 (Von Neumann–Morgenstern representation). *Let X be finite. If a preference relation $\succsim$ on $\Delta(X)$ satisfies completeness, transitivity, continuity, and independence, then there exists a utility index $u : X \rightarrow \mathbb{R}$ such that*

$$U(p) = \sum_{x \in X} p(x)u(x)$$

represents $\succsim$ on $\Delta(X)$:

$$p \succsim q \iff U(p) \geq U(q).$$

*The index u is unique up to positive affine transformations.*¹⁸

Remark (Geometry on the three-outcome simplex). Let $X = \{x_1, x_2, x_3\}$ and write $p_i = p(x_i)$, with

$$p_1 + p_2 + p_3 = 1.$$

If $u_i := u(x_i)$, then

$$U(p) = p_1 u_1 + p_2 u_2 + p_3 u_3 = (u_1 - u_3)p_1 + (u_2 - u_3)p_2 + u_3.$$

Thus, in the (p_1, p_2) simplex,

$$\{(p_1, p_2) : p_1 \geq 0, p_2 \geq 0, p_1 + p_2 \leq 1\},$$

indifference sets are straight and parallel lines:

$$(u_1 - u_3)p_1 + (u_2 - u_3)p_2 = \bar{U} - u_3.$$

When $u_2 \neq u_3$, their slope is

$$-\frac{u_1 - u_3}{u_2 - u_3}.$$

Upper contour sets are half-planes intersected with the simplex, hence convex.

¹⁷ *Intuition for future me:* If you prefer p to q , then you should also prefer any mixture of p with a third lottery r to the same mixture of q with r .

On the lottery space $\Delta(X)$, completeness + transitivity + (topological) continuity already guarantee the existence of some utility $U : \Delta(X) \rightarrow \mathbb{R}$ representing $\succsim$. Independence is not needed for existence; it is the axiom that forces U to be affine in probabilities (vNM E.U.). With Independence, there is a cardinal index $u : X \rightarrow \mathbb{R}$ with $U(p) = \sum_{x \in X} p(x)u(x)$, unique up to positive affine transformations.

¹⁸ *Intuition for future me:* Preferences over lotteries can be represented as the expected value of a utility index defined over deterministic outcomes. Independence is the axiom that gives linearity in probabilities.

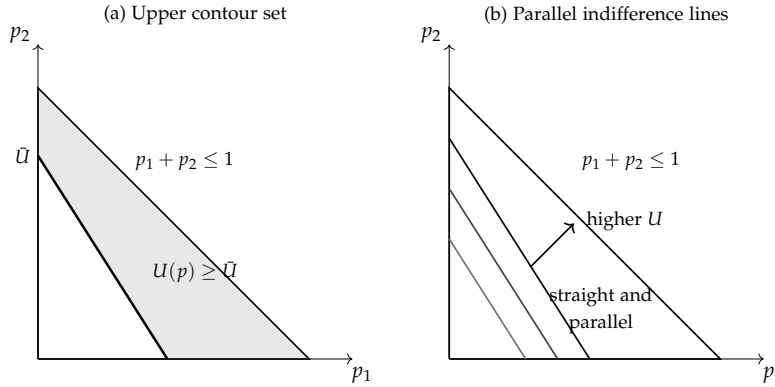


Figure 19: Von Neumann–Morgenstern geometry on the three-outcome simplex. Left: an upper contour set is a half-plane intersected with the simplex. Right: because expected utility is affine in probabilities, indifference sets are straight and parallel.

Normalization and binary reduction

Because X is finite, there exist a *best* and a *worst* outcome w.r.t. $\succsim$: pick $\bar{x} \in X$ with $\delta_{\bar{x}} \succsim \delta_x$ for all x , and $\underline{x} \in X$ with $\delta_x \succsim \delta_{\underline{x}}$ for all x . Normalize the VNM index by¹⁹

$$u(\underline{x}) = 0, \quad u(\bar{x}) = 1.$$

Then for any lottery $p \in \Delta(X)$,

$$U(p) = \sum_{x \in X} p(x) u(x) \in [0, 1],$$

and, by affinity,

$$p \sim U(p) \delta_{\bar{x}} + (1 - U(p)) \delta_{\underline{x}}.$$

Remark. Writing $\alpha p + (1 - \alpha)q$ componentwise gives

$$\sum_{x \in X} (\alpha p(x) + (1 - \alpha)q(x)) u(x) = \alpha \sum_{x \in X} p(x) u(x) + (1 - \alpha) \sum_{x \in X} q(x) u(x),$$

i.e., the indifference sets are level lines of a linear form in $(p(x))_{x \in X}$, which is exactly what the simplex plots above are visualizing.

¹⁹ *Intuition for future me:* Every lottery is *indifferent* to a two-point lottery between the best and worst outcomes, with weight on the best equal to its expected utility. Thus ranking lotteries is the same as ranking these binary mixtures by their weight on $\bar{x}$.

Class 9 – Objective vs. Subjective Probabilities

In the theory of choice under uncertainty we can distinguish two versions:

- (a) **Objective probabilities:** true probabilities for each outcome are given.
- (b) **Subjective probabilities:** probabilities are not given; instead, from the choices of the agent we attempt to infer beliefs.

Objective probabilities and the VNM framework

Let X be a finite set of outcomes. Then $\Delta(X)$ denotes the set of all probability distributions over X , that is, all vectors $p = (p(x))_{x \in X}$ with $\sum_{x \in X} p(x) = 1$.

A preference relation $\succsim$ is defined over $\Delta(X)$, and we impose the following axioms:

- (i) **Rationality (R).** $\succsim$ is complete and transitive.
- (ii) **Continuity (C).** $\succsim$ is continuous, hence admits a continuous utility representation.
- (iii) **Independence (I).** For all $p, q, r \in \Delta(X)$ and $\alpha \in (0, 1)$,

$$p \succsim q \implies \alpha p + (1 - \alpha)r \succsim \alpha q + (1 - \alpha)r.$$

Theorem 5 (Von Neumann–Morgenstern Representation). *If a preference relation $\succsim$ on $\Delta(X)$ satisfies (R), (C), and (I), then there exists a function $u : X \rightarrow \mathbb{R}$ such that for all $p, q \in \Delta(X)$,*

$$p \succsim q \iff \sum_{x \in X} p(x)u(x) \geq \sum_{x \in X} q(x)u(x).$$

Proof sketch. Part 1. Construct $U(p) = \sum_{x \in X} p(x)u(x)$ and show that this is well defined under (R), (C), (I).

Part 2. Show that $U(p)$ correctly orders the elements of $\Delta(X)$, i.e. $p \succsim q$ iff $U(p) \geq U(q)$. $\square$

Remark (Strict separability). The independence axiom induces a strong form of separability: the utility of a mixture is the corresponding mixture of utilities. This is exactly what makes expected utility a *linear* functional of probabilities.

Remark (Risk neutrality baseline). The expected utility representation itself is agnostic about attitudes toward risk. Neutrality is the natural baseline. Risk aversion or risk seeking only appear once we apply u to monetary outcomes and study the curvature of u .

Rewriting lotteries

Given $p \in \Delta(X)$, we can always write it as a convex combination of degenerate lotteries:

$$p = \sum_{x \in X} p(x) \delta_x.$$

This allows us to rewrite rankings in terms of comparisons of degenerate lotteries weighted by probabilities. For instance, with three outcomes:

$$\begin{pmatrix} p(x) \\ p(y) \\ p(z) \end{pmatrix} = p(x) \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + p(y) \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + p(z) \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

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²⁰ *Intuition for future me:* The key is that rankings over lotteries should depend on whether probabilities differ. Expected utility treats these mixtures in a linear fashion.

Normalization and affine transformations

Suppose $u : X \rightarrow \mathbb{R}$ is a VNM index representing $\succsim$ through

$$U(p) = \sum_{z \in X} p(z)u(z), \quad p \in \Delta(X).$$

That is,

$$p \succsim q \iff U(p) \geq U(q).$$

Theorem 6 (Uniqueness of the VNM index). *Let $u, v : X \rightarrow \mathbb{R}$ be two VNM indices representing the same preference relation $\succsim$ on $\Delta(X)$, so that*

$$U(p) = \sum_{z \in X} p(z)u(z), \quad V(p) = \sum_{z \in X} p(z)v(z)$$

both represent $\succsim$. Then there exist constants $a > 0$ and $b \in \mathbb{R}$ such that

$$v(z) = a u(z) + b \quad \text{for all } z \in X.$$

Remark. The affine restriction is specific to VNM representations. An arbitrary ordinal representation of $\succsim$ can be transformed by any strictly increasing function, but a representation that remains linear in probabilities can only be changed by a positive affine transformation.

Remark (Expected utility as probability weights). Expected utility assigns probabilities to outcomes in such a way that makes the agent indifferent between compound and reduced lotteries. This is the formal content of the independence axiom.

The Allais Paradox: when Independence fails

Example (Allais paradox). Consider three monetary outcomes

$$27,000 \succ 24,000 \succ 0.$$

The Allais paradox compares the following two choice problems.

First choice problem.

$$\begin{aligned} A : 0.33 \rightarrow 27,000, \quad 0.66 \rightarrow 24,000, \quad 0.01 \rightarrow 0, \\ B : 1 \rightarrow 24,000. \end{aligned}$$

Many agents choose the sure lottery:

$$B \succ A.$$

Second choice problem.

$$\begin{aligned} A' : 0.33 \rightarrow 27,000, \quad 0.67 \rightarrow 0, \\ B' : 0.34 \rightarrow 24,000, \quad 0.66 \rightarrow 0. \end{aligned}$$

Many agents choose the lottery with the higher upside:

$$A' \succ B'.$$

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Remark (Why this violates expected utility). Normalize the VNM index by

$$u(0) = 0, \quad u(27,000) = 1, \quad u(24,000) = x.$$

If expected utility represents preferences, then

$$U(A) = 0.33 + 0.66x, \quad U(B) = x.$$

Thus

$$B \succ A \iff x > 0.33 + 0.66x \iff x > \frac{33}{34}.$$

On the other hand,

$$U(A') = 0.33, \quad U(B') = 0.34x.$$

Thus

$$A' \succ B' \iff 0.33 > 0.34x \iff x < \frac{33}{34}.$$

²¹ *Intuition for future me:* The first choice displays a certainty effect: the sure payoff 24,000 is attractive enough to beat a lottery with a slightly higher prize. But once the common safe component is removed, many people switch toward the lottery with a chance of 27,000. This switch violates Independence.

The two observed choices therefore impose contradictory restrictions on x . Hence the pattern

$$B \succ A \quad \text{and} \quad A' \succ B'$$

cannot be represented by expected utility.

Remark (Common-consequence interpretation). Define the residual lotteries

$$C := \frac{33}{34} \delta_{27,000} + \frac{1}{34} \delta_0, \quad D := \delta_{24,000}.$$

Then the first pair can be written as

$$A = 0.34 C + 0.66 \delta_{24,000}, \quad B = 0.34 D + 0.66 \delta_{24,000}.$$

The second pair can be written as

$$A' = 0.34 C + 0.66 \delta_0, \quad B' = 0.34 D + 0.66 \delta_0.$$

Thus both choice problems compare the same underlying lotteries C and D , but mixed with different common consequences. Independence says that adding the same common consequence should not reverse the ranking. Therefore,

$$B \succ A \implies D \succ C \implies B' \succ A',$$

contradicting the observed Allais pattern $A' \succ B'$.

Class 10 – Subjective Probabilities

Motivation

We aim to infer *subjective beliefs* from choices. The key shift in this part is that the *objects of choice are acts*—state-contingent plans—not bare outcomes or exogenously risky lotteries. If you repeatedly prefer the act “\$100 if A , \$0 otherwise” to the act “\$100 if B , \$0 otherwise,” holding consequences fixed, your choices reveal that you *regard event A as more likely than B* . Preferences over acts thus encode both *tastes* over consequences and *beliefs* about states.

Roadmap. (1) Define *states*, *events*, *consequences*, and *acts* (functions from states to consequences). (2) Take preferences over acts as primitive. (3) Impose behavioral axioms (including the Sure-Thing Principle) under which preferences admit a representation as *expected utility* with a *subjective probability* on events.

Intuition: An act is an *if-then* rule. We keep the *consequence menu* fixed and vary the *event* that triggers it. The way you rank such rules reveals a likelihood ordering over events.

Framework

Definition (States, events, acts, and consequences). Let S be a (finite) set of *states of the world* and let $\mathcal{E} \subseteq 2^S$ be the set of *events*. Let X be a set of *consequences*. An *act* is a function $a : S \rightarrow X$ that assigns a consequence to each state. A (complete and transitive) preference relation $\succeq$ is defined over the set of acts $\mathcal{A} = X^S$.

Remark (Why acts are the primitives here). Unlike in objective-risk problems (where lotteries are given with known probabilities), here probabilities are *not* assumed; they are inferred from $\succeq$ over acts. Acts are the natural primitives because they tell us, for each possible state, *what would happen if this plan were chosen*. Events matter only through how acts condition consequences on them.

Remark (When consequences are monetary). In many applications we take $X = [a, b] \subset \mathbb{R}$ (money or consumption) and we already endowed X with a vNM utility index $u : X \rightarrow \mathbb{R}$. We also write $\Delta(X)$ for the set

of *simple lotteries* over X (probability distributions with finite support).

Remark (Important: timing). We must choose an act $a \in \mathcal{A}$ *before* learning which state $s \in S$ will prevail. After the state realizes, the chosen act mechanically yields its corresponding consequence $a(s)$.

Representation: subjective expected utility (SEU)

The idea is that preferences over acts can be represented as expected utility with respect to a *subjective* probability on states.

Theorem 7 (SEU representation (finite S ; Anscombe–Aumann flavor)). Suppose $\succeq$ over $\mathcal{A}$ satisfies the usual list of axioms:²² then there exist (i) a probability distribution p on S and (ii) a utility index $u : X \rightarrow \mathbb{R}$ such that for all acts $a, b \in \mathcal{A}$,

$$a \succeq b \iff \sum_{s \in S} p(s) u(a(s)) \geq \sum_{s \in S} p(s) u(b(s)).$$

Moreover, p is unique, and u is unique up to a positive affine transformation $u'(x) = \alpha u(x) + \beta$ with $\alpha > 0$.

Remark (Why the Sure-Thing Axiom matters). The Sure-Thing Principle (STP) says: if two acts deliver the *same* consequence on an event E , then your ranking between them should depend only on what they do on E^c . In SEU, this follows from linearity of expectation and is **equivalent to independence** across events.

Remark (Why the Sure-Thing Principle matters). The Sure-Thing Principle says that common consequences should not affect choice. If two acts deliver the same consequence c on an event E , then under SEU,

$$EU(f) = \Pr(E)u(c) + \Pr(E^c)EU(f | E^c),$$

$$EU(g) = \Pr(E)u(c) + \Pr(E^c)EU(g | E^c).$$

The common term $\Pr(E)u(c)$ cancels out, so the ranking between f and g depends only on what they imply on E^c .²³

Add exercise 3 of PS5b as an example.

Ellsberg's paradox: a failure of STP (ambiguity)

Consider an urn with 90 balls: 30 are red (R); the remaining 60 are blue (B) or green (G) in unknown proportions (ambiguity about B vs. G). The state space is $S = \{R, B, G\}$. Pay \$100 if the described event occurs, and \$0 otherwise.

²² Weak order, continuity, monotonicity w.r.t. consequences, and the Sure-Thing/independence axiom across events; plus nondegeneracy. In the AA setup, mixture independence is imposed via objective mixtures of consequences.

Reading: "Preferences reveal your subjective probabilities, and they admit an expected-utility representation."

²³ *Intuition for future me*: If two acts coincide on E , then E is irrelevant for comparing them; only the part where they differ matters.

Act	R	B	G
A	100	0	0
B	0	100	0
A'	100	0	100
B'	0	100	100

Example (Typical choices). Empirically, many subjects choose $A \succ B$ (bet on red) and also $B' \succ A'$ (bet on blue-or-green rather than red-or-green).

Remark (Why this contradicts SEU / STP). Observe that A' and B' are obtained from A and B by adding the *same* payoff on event G (both give \$100 on G). By the Sure-Thing Principle, adding a common consequence on G should not flip the ranking: from $A \succ B$ we should infer $A' \succ B'$. The observed reversal $B' \succ A'$ therefore violates STP (and hence SEU). This is interpreted as *ambiguity aversion*: people dislike bets whose probabilities are ill-defined (here, the split between B and G).

Connection to “probabilistic sophistication”: if preferences depended only on the induced lotteries over consequences, a single subjective p would rationalize both choices. Ellsberg shows they often do not.

Event comparisons via bets

Fix consequences and vary only the event that triggers the prize. For example, compare the two acts

$$a_E(s) = \begin{cases} x_H & \text{if } s \in E, \\ x_L & \text{if } s \notin E, \end{cases} \quad a_F(s) = \begin{cases} x_H & \text{if } s \in F, \\ x_L & \text{if } s \notin F, \end{cases}$$

with $x_H \succ x_L$. If $a_E \succeq a_F$ for all such pairs (x_H, x_L) , we read this as “ E is (subjectively) at least as likely as F ,” which SEU rationalizes via $p(E) \geq p(F)$.

Utility for money and simple lotteries (Chapter 6)

We now specialize to monetary consequences. Let $X = [a, b] \subset \mathbb{R}$. Denote by $\Delta(X)$ the set of *simple lotteries* on X , i.e. probability distributions with finite support.

Definition (vNM index on money). A preference $\succeq$ on $\Delta(X)$ has a vNM representation if there exists $u : X \rightarrow \mathbb{R}$ such that for all $\pi, \rho \in \Delta(X)$,

$$\pi \succeq \rho \iff \sum_{x \in \text{supp}(\pi)} \pi(x) u(x) \geq \sum_{x \in \text{supp}(\rho)} \rho(x) u(x).$$

The function u is unique up to positive affine transformations.

Definition (Monotonicity on money). Write $\delta_x \in \Delta(X)$ for the degenerate lottery that pays x for sure. A preference $\succeq$ is *increasing* if for all $x, y \in [a, b]$, $x \geq y$ implies $\delta_x \succeq \delta_y$ (strictly if $x > y$).

Remark. Intuition: “More money is (weakly) better for sure outcomes.” In a vNM representation, this is equivalent to u being (weakly) increasing on $[a, b]$.

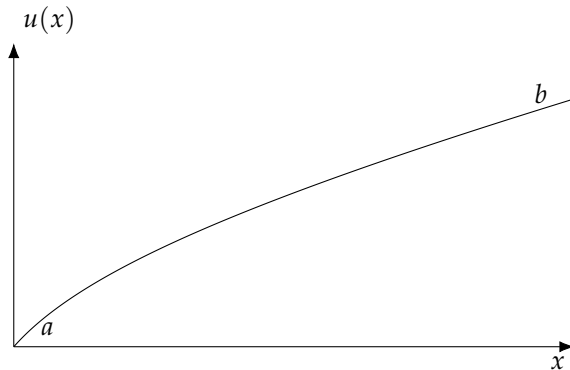


Figure 20: An increasing vNM index u on money.

First-order stochastic dominance (FOSD)

Definition (FOSD via CDFs). For $\pi, \rho \in \Delta(X)$ with cumulative distribution functions F_π and F_ρ , we say that π *first-order stochastically dominates* ρ (write $\pi \succeq_{\text{FOSD}} \rho$) if

$$F_\pi(x) \leq F_\rho(x) \quad \text{for all } x \in [a, b],$$

with strict inequality for some x (for strict FOSD). If the CDFs cross, the pair is not comparable by FOSD.

Proposition (Monotonicity $\iff$ FOSD). Let $\succeq$ on $\Delta(X)$ admit a vNM representation by u . Then

$$\succeq \text{ is increasing} \iff u \text{ is (weakly) increasing} \iff (\forall \pi, \rho \in \Delta(X)) \pi \succeq_{\text{FOSD}} \rho \Rightarrow \pi \succeq \rho.$$

Proof sketch. ($\Rightarrow$) If $\delta_x \succeq \delta_y$ for $x \geq y$, then $u(x) \geq u(y)$, so u is increasing; FOSD implies higher probability on higher payoffs, and linearity of EU with increasing u gives $\pi \succeq \rho$. ($\Leftarrow$) If u were not increasing, there exist $x > y$ with $u(x) < u(y)$: but $\delta_x \succeq_{\text{FOSD}} \delta_y$ while EU would rank $\delta_y \succ \delta_x$, a contradiction.

Reading tip. If $F_\pi \leq F_\rho$, every increasing u yields $\mathbb{E}_\pi[u(X)] \geq \mathbb{E}_\rho[u(X)]$. If CDFs cross, some increasing u prefers π and another prefers ρ .

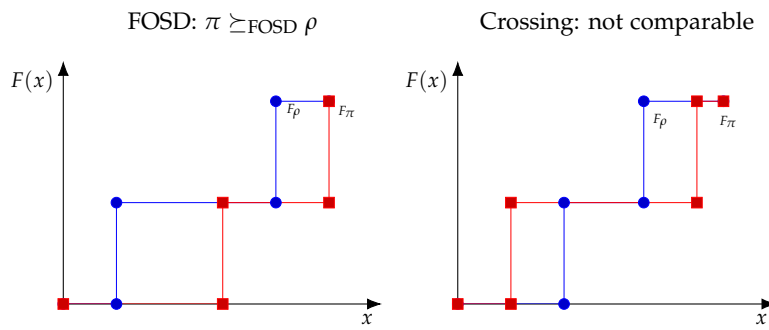


Figure 21: Left: $F_\pi \leq F_\rho$ pointwise (FOSD). Right: CDFs cross $\Rightarrow$ no FOSD ranking.

Class 11 – Utility of Money (Chapter 6)

Setup and simple lotteries

Let $X = [0, \bar{x}] \subset \mathbb{R}$ denote monetary outcomes, and let $\Delta(X)$ be the set of *simple lotteries* (finite support probability distributions) over X . Preferences $\succeq$ are defined over $\Delta(X)$.

Definition (EU representation on money). A vNM index $u : X \rightarrow \mathbb{R}$ represents $\succeq$ if for all $\pi, \rho \in \Delta(X)$,

$$\pi \succeq \rho \iff \underbrace{U(\pi)}_{:= \mathbb{E}_\pi[u(X)]} = \sum_{x \in \text{supp}(\pi)} \pi(x) u(x) \geq \sum_{x \in \text{supp}(\rho)} \rho(x) u(x) = U(\rho),$$

with u unique up to positive affine transformations.

Monotonicity and FOSD

Definition (Monotonicity on sure amounts). Let δ_x be the degenerate lottery that pays x for sure. We say that $\succeq$ is *increasing* if for all $x \geq y$ in X , $\delta_x \succeq \delta_y$ (strictly increasing if $x > y \Rightarrow \delta_x \succ \delta_y$).

Proposition (Three equivalent statements). For preferences $\succeq$ on $\Delta(X)$ with EU index u , the following are equivalent:

- (i) $\succeq$ is (weakly) increasing.
- (ii) u is (weakly) increasing on $[0, \bar{x}]$.
- (iii) If π first-order stochastically dominates ρ (i.e. $F_\pi(x) \leq F_\rho(x)$ for all x), then $\pi \succeq \rho$.

Moreover, if $\succeq$ is strictly increasing, then u is strictly increasing and strict FOSD implies strict preference ($F_\pi \leq F_\rho$ and $F_\pi < F_\rho$ somewhere $\Rightarrow \pi \succ \rho$).

Risk aversion and SOSD

Definition (Risk aversion). A preference relation $\succeq$ on $\Delta(X)$ is *risk-averse* if for every lottery $\pi \in \Delta(X)$,

$$\delta_{\mathbb{E}_\pi[X]} \succeq \pi.$$

That is, the sure payoff equal to the lottery's expected value is weakly preferred to the lottery itself. Strict risk aversion means that the preference is strict whenever π is non-degenerate.

Proposition (Characterizations of risk aversion). *For EU preferences with index u , the following are equivalent:*

- (a) $\succeq$ is (strictly) risk-averse.
- (b) u is (strictly) concave on $[0, \bar{x}]$.
- (c) (SOSD consistency) If π second-order stochastically dominates ρ (write $\pi \succeq_{\text{SOSD}} \rho$), then $\pi \succeq \rho$. With strict concavity, $\pi \succ \rho$ when the dominance is strict.

Sketch. (a) $\Leftrightarrow$ (b) is Jensen: $u(\mathbb{E}[X]) \geq \mathbb{E}[u(X)]$, with $>$ if u is strictly concave and π non-degenerate. (b) $\Rightarrow$ (c): concavity plus the definition of SOSD (below) implies $\mathbb{E}_\pi[u(X)] \geq \mathbb{E}_\rho[u(X)]$. (c) $\Rightarrow$ (a) by taking $\rho = \pi$ and $\pi = \delta_{\mathbb{E}[X]}$, which SOSD-dominates π . $\square$

Definition (Second-order stochastic dominance). Let $X = [0, \bar{x}]$. For $\pi, \rho \in \Delta(X)$ with CDFs F_π, F_ρ , we write

$$\pi \succeq_{\text{SOSD}} \rho \iff \int_0^t F_\pi(x) dx \leq \int_0^t F_\rho(x) dx \text{ for all } t \in [0, \bar{x}],$$

and

$$\int_0^{\bar{x}} F_\pi(x) dx = \int_0^{\bar{x}} F_\rho(x) dx \quad (\text{equivalently, } \mathbb{E}_\pi[X] = \mathbb{E}_\rho[X]).$$

Remark (Mean-preserving spread (MPS)). ρ is an MPS of π iff there exists a zero-mean noise $\tilde{Y}$ such that

$$X_\rho = X_\pi + \tilde{Y}, \quad \mathbb{E}[\tilde{Y} | X_\pi] = 0.$$

This is equivalent to $\pi \succeq_{\text{SOSD}} \rho$. Intuition: ρ is obtained by “spreading” mass of π away from its mean without changing the mean.

Takeaways

- Monotonicity on sure amounts $\Leftrightarrow u$ increasing $\Leftrightarrow$ FOSD consistency (Prop.).
- Risk aversion $\Leftrightarrow u$ concave $\Leftrightarrow$ SOSD consistency (Prop.).
- Mean-preserving spreads (zero-mean noise added) worsen lotteries for a risk-averse DM; they leave the mean unchanged but raise dispersion.

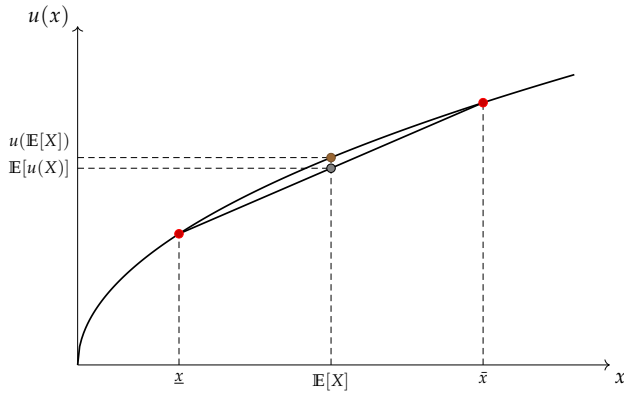


Figure 22: Risk aversion: if u is concave, then Jensen's inequality implies

$$u(\mathbb{E}[X]) \geq \mathbb{E}[u(X)].$$

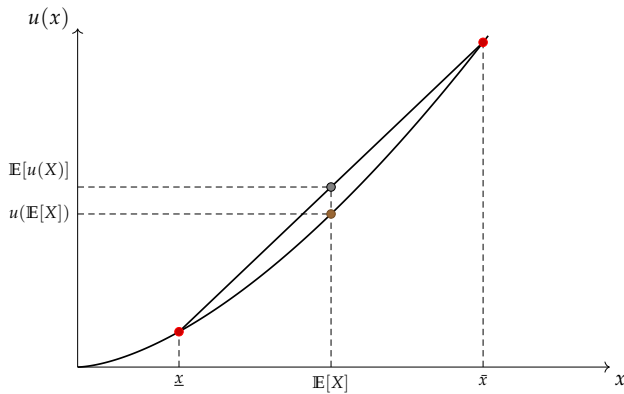


Figure 23: Risk loving: if u is convex, then Jensen's inequality is reversed:

$$u(\mathbb{E}[X]) \leq \mathbb{E}[u(X)].$$

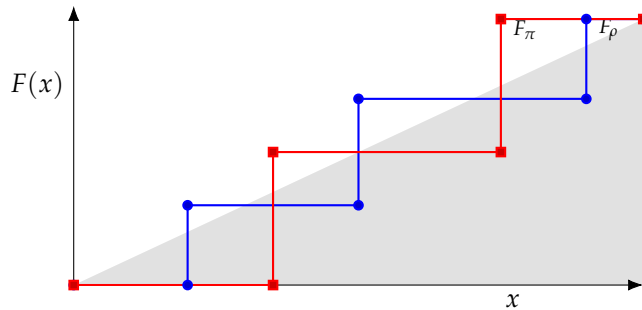


Figure 24: Second-order dominance: $\int_0^t F_\pi \leq \int_0^t F_\rho$ for all t (same mean). Then every increasing concave u prefers π to ρ .

Arrow–Pratt, certainty equivalent, and risk premium

Definition (Arrow–Pratt absolute risk aversion). For an EU index $u : X \rightarrow \mathbb{R}$ with $u' > 0$, the *absolute risk-aversion* (ARA) coefficient is

$$A_u(x) := -\frac{u''(x)}{u'(x)}.$$

It is *invariant to positive affine transformations*: if $v = \alpha u + \beta$ with $\alpha > 0$, then $A_v \equiv A_u$.

Definition (Certainty equivalent and risk premium). Given a lottery

$\pi \in \Delta(X)$, the *certainty equivalent* $CE(\pi) \in X$ is the (unique) number such that

$$u(CE(\pi)) = \mathbb{E}_\pi[u(X)].$$

The *risk premium* is

$$RP(\pi) := \mathbb{E}_\pi[X] - CE(\pi).$$

Proposition (Risk aversion and nonnegative risk premia). *For expected-utility preferences with u increasing, the following are equivalent:*

- (i) *The decision maker is (weakly) risk-averse.*
- (ii) *The Bernoulli utility u is (weakly) concave.*
- (iii) *For every lottery π , the certainty equivalent satisfies*

$$CE(\pi) \leq \mathbb{E}_\pi[X].$$

Equivalently, the risk premium is nonnegative.

Sketch. By Jensen's inequality, concavity implies

$$u(\mathbb{E}[X]) \geq \mathbb{E}[u(X)].$$

Since the certainty equivalent CE is defined by

$$u(CE) = \mathbb{E}[u(X)],$$

and u is increasing, it follows that

$$CE \leq \mathbb{E}[X] \Rightarrow RP = \mathbb{E}[X] - CE \geq 0.$$

Under strict concavity, the inequalities are strict for every nondegenerate lottery. $\square$

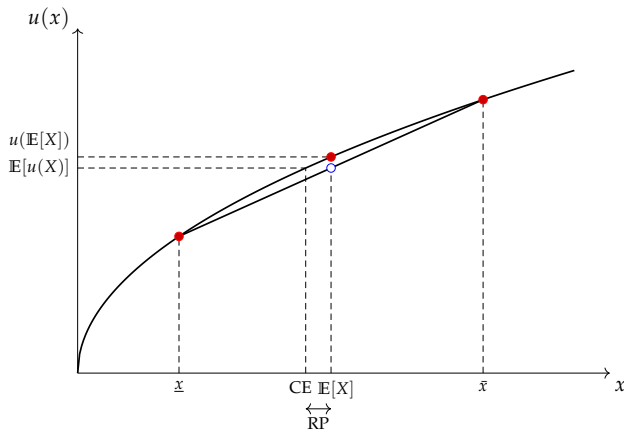


Figure 25: For a risk-averse expected-utility maximizer, concavity implies

$$u(\mathbb{E}[X]) \geq \mathbb{E}[u(X)].$$

The certainty equivalent is defined by $u(CE) = \mathbb{E}[u(X)]$, so

$$CE \leq \mathbb{E}[X].$$

Hence the risk premium is nonnegative:

$$RP = \mathbb{E}[X] - CE \geq 0.$$

Small-risk expansion and Arrow–Pratt

Fix $x \in X$ and let $\tilde{Z}$ be a mean-zero r.v. with $\mathbb{E}[\tilde{Z}] = 0$ and $\mathbb{E}[\tilde{Z}^2] < \infty$. For $\varepsilon \geq 0$, define the **risk premium** $c(\varepsilon)$ by the CE equation

$$u(x - c(\varepsilon)) = \mathbb{E}[u(x + \varepsilon \tilde{Z})].$$

Then $c(0) = 0$, $c'(0) = 0$, and if $u \in C^2$ and c is twice differentiable at 0,

$$c''(0) = A_u(x) \mathbb{E}[\tilde{Z}^2] = -\frac{u''(x)}{u'(x)} \mathbb{E}[\tilde{Z}^2]$$

so that, for small risks,

$$\text{RP}(\varepsilon \tilde{Z}) = c(\varepsilon) \approx \frac{1}{2} A_u(x) \varepsilon^2 \text{Var}(\tilde{Z}).$$

Derivation. Apply a second-order Taylor expansion to both sides around $\varepsilon = 0$ and use $\mathbb{E}[\tilde{Z}] = 0$.

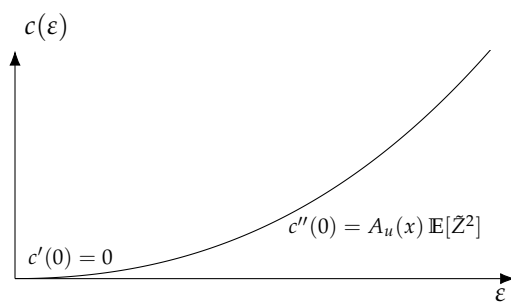


Figure 26: Small-risk premium $c(\varepsilon)$: tangent flat at 0 and convex when $A_u(x) > 0$.

Constant absolute risk aversion (CARA)

Definition (CARA). u has constant ARA equal to $\rho > 0$ if $A_u(x) \equiv \rho$ for all x , i.e.

$$-\frac{u''(x)}{u'(x)} = \rho \iff u''(x) + \rho u'(x) = 0.$$

Solving the ODE yields $u(x) = \alpha - \beta e^{-\rho x}$ with $\beta > 0$. A convenient normalization is $u(x) = 1 - e^{-\rho x}$.

Proposition (Comparative risk aversion (Pratt)). If $A_u(x) \geq A_v(x)$ for all x , then u is uniformly more risk-averse than v (equivalently, $v = \phi \circ u$ for some increasing concave ϕ). Hence $\text{RP}_u(\pi) \geq \text{RP}_v(\pi)$ for all π .

A simple portfolio choice “at zero”

Let initial wealth be $m > 0$. Choose a share $\lambda \in [0, 1]$ to invest in a risky increment $\tilde{X}$ while the remainder stays safe. Final wealth:

$$W(\lambda) = (1 - \lambda) m + \lambda \tilde{X}.$$

The DM chooses λ to maximize $E[u(W(\lambda))]$.

Proposition (Sign of the optimal direction at $\lambda = 0$). *Suppose u is increasing and differentiable. Then*

$$\left. \frac{d}{d\lambda} \mathbb{E}[u(W(\lambda))] \right|_{\lambda=0} = u'(m) (\mathbb{E}[\tilde{X}] - m).$$

Hence, if an optimum is interior near 0, we have $\lambda^* > 0$ iff $\mathbb{E}[\tilde{X}] > m$. If $\tilde{X}$ is a net risky payoff over the safe (so the safe is 0), this simplifies to $\lambda^* > 0$ iff $\mathbb{E}[\tilde{X}] > 0$.

Remark. For (small) mean-zero risks with variance σ^2 , the second-order effect is governed by Arrow–Pratt:

$$\left. \frac{d^2}{d\lambda^2} \mathbb{E}[u(W(\lambda))] \right|_{\lambda=0} = u'(m) A_u(m) \sigma^2 (-\text{sign}),$$

so risk aversion ($A_u > 0$) penalizes variance and pushes the optimum λ^* downward.

Class 12 – Theory of the Firm (Chapter 9)

Setup

We model the firm as a single decision maker. The *technology* is a production set

$$\mathcal{Z} \subseteq \mathbb{R}^k,$$

whose elements $z = (z_1, \dots, z_k)$ are *production plans*. We follow the sign convention:

$$z_i \leq 0 \text{ (inputs),} \quad z_j \geq 0 \text{ (outputs).}$$

Example. With two inputs and one output, $z = (-2, -3, 24) \in \mathcal{Z}$ means the firm uses 2 units of input 1 and 3 of input 2 to produce 24 units of output.

Remark. This sign convention makes the language flexible: “adding output” increases components of z that are outputs, while “using more inputs” decreases the corresponding (negative) components.

Prices and profits

Let $p \in \mathbb{R}^k$ be the *price vector*. The firm’s (static) profit from plan z is

$$\pi(p, z) = p \cdot z = \sum_{i=1}^k p_i z_i.$$

With outputs priced $p_j \geq 0$ and inputs priced $p_i \geq 0$ (remember $z_i \leq 0$ for inputs), outputs raise profit and inputs reduce it.

The firm’s problem and the profit function

Given p , the set of profit-maximizing plans is

$$Z(p) := \arg \max_{z \in \mathcal{Z}} p \cdot z,$$

which can be a *correspondence* (multiple maximizers) or empty. The associated profit function is

$$\pi(p) := \sup \{ p \cdot z : z \in \mathcal{Z} \} \quad (\text{well-defined once } \mathcal{Z} \neq \emptyset),$$

and may take value $+\infty$ if technology allows arbitrarily profitable scale at prices p .

Remark (Existence (minimal conditions)). If $\mathcal{Z}$ is nonempty, closed, and profits are *bounded above* at p (no arbitrarily scalable free output in a priced direction), then $Z(p) \neq \emptyset$. Uniqueness generally needs strict concavity of the feasible frontier (or of π).

A two-dimensional picture

Consider one input $L \leq 0$ and one output $y \geq 0$. A convenient representation is

$$\mathcal{Z} = \{(L, y) \in \mathbb{R}^2 : y \leq \sqrt{-L}, L \leq 0, y \geq 0\},$$

which says higher output requires (weakly) more input. The feasible set is the area under the curve $y = \sqrt{-L}$ for $L \leq 0$.

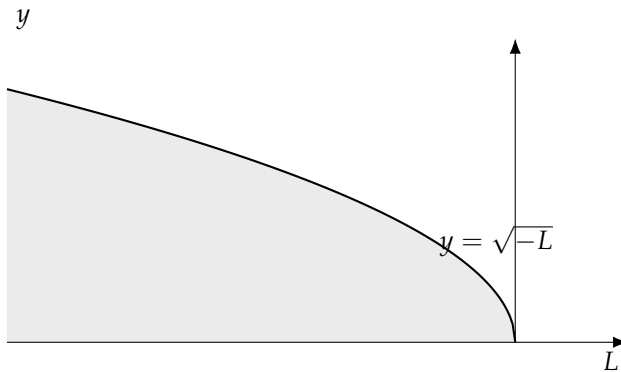


Figure 27: Technology with one input ($L \leq 0$) and one output ($y \geq 0$). The shaded area is the feasible set $\mathcal{Z}$.

Iso-profit lines. At prices (p_L, p_y) with $p_L, p_y > 0$, iso-profit lines in (L, y) -space have slope $-\frac{p_L}{p_y}$ (since $\pi = p_y y + p_L L$ with $L \leq 0$). Profit maximization picks a boundary point of $\mathcal{Z}$ where an iso-profit line is just tangent (or a corner like the origin if prices are unfavorable).

A warm-up example (CRS ray and (non)existence)

Let $k = 2$ with $z = (z_1, z_2)$, where $z_1 \leq 0$ is an input and $z_2 \geq 0$ an output. Consider the constant-returns technology

$$\mathcal{Z} = \{(z_1, z_2) \in \mathbb{R}^2 : z_1 \leq 0, z_2 = |z_1|\} = \{(-t, t) : t \geq 0\}.$$

At prices $p = (p_1, p_2)$ the firm solves $\max_{t \geq 0} p \cdot (-t, t) = t(p_2 - p_1)$. Hence

$$Z(p) = \begin{cases} \{(0, 0)\} & \text{if } p_1 > p_2, \\ \{(-t, t) : t \geq 0\} & \text{if } p_1 = p_2, \\ \emptyset & \text{if } p_1 < p_2, \end{cases} \quad \pi(p) = \begin{cases} 0 & \text{if } p_1 \geq p_2, \\ +\infty & \text{if } p_1 < p_2. \end{cases}$$

Lesson: if an output's price exceeds the "cost" of the input along a CRS ray, profits are unbounded and no maximizer exists.

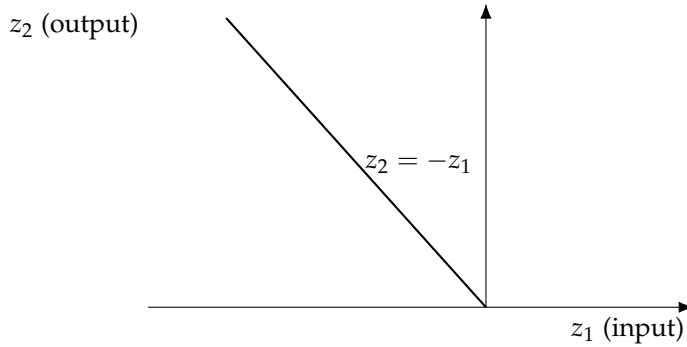


Figure 28: CRS ray. If $p_1 > p_2$, the maximizer is $(0,0)$; if $p_1 < p_2$, profits are unbounded.

Existence, continuity, and convexity of the choice set

Proposition (Existence under compactness). *If $\mathcal{Z}$ is nonempty and compact, then for every p the maximization $\max_{z \in \mathcal{Z}} p \cdot z$ admits a solution; i.e. $Z(p) \neq \emptyset$.*

Remark (Free disposal as a fallback). If $0 \in \mathcal{Z}$ and the technology has *free disposal* (i.e. $z' \leq z$ componentwise and $z \in \mathcal{Z}$ imply $z' \in \mathcal{Z}$), then for any $p \geq 0$ the problem has at least the feasible plan $z = 0$ with $\pi(p, 0) = 0$.

Theorem 8 (Berge's Maximum Theorem). *If $\mathcal{Z}$ is nonempty and compact, then the argmax correspondence $Z(\cdot)$ is upper hemicontinuous (hence has a closed graph), and the profit function $\pi(p) = \max_{z \in \mathcal{Z}} p \cdot z$ is continuous in p .*

Proposition (Convex choice at fixed prices). *If $\mathcal{Z}$ is convex, then $Z(p)$ is convex for every p .*

Proof. If $Z(p) = \emptyset$, then $Z(p)$ is convex by convention. Suppose instead that $z, z' \in Z(p)$ and let $\lambda \in [0, 1]$. Define

$$\tilde{z} := \lambda z + (1 - \lambda)z'.$$

Since $\mathcal{Z}$ is convex and $z, z' \in \mathcal{Z}$, we have

$$\tilde{z} \in \mathcal{Z}.$$

Moreover, because $z, z' \in Z(p)$, both attain the maximal profit:

$$p \cdot z = \pi(p) \quad \text{and} \quad p \cdot z' = \pi(p).$$

Therefore, by linearity of the objective,

$$p \cdot \tilde{z} = p \cdot (\lambda z + (1 - \lambda)z') = \lambda p \cdot z + (1 - \lambda)p \cdot z' = \lambda \pi(p) + (1 - \lambda)\pi(p) = \pi(p).$$

Thus $\tilde{z}$ is feasible and also attains the maximum profit. Hence

$$\tilde{z} \in Z(p).$$

So $Z(p)$ is convex. □

Homogeneity and the law of supply (WAPM)

Proposition (Positive homogeneity). *For all $\lambda > 0$ and all p ,*

$$Z(\lambda p) = Z(p) \quad \text{and} \quad \pi(\lambda p) = \lambda \pi(p).$$

Proposition (Weak axiom of profit maximization / law of supply). *If $z \in Z(p)$ and $z' \in Z(p')$, then*

$$(z - z') \cdot (p - p') \geq 0.$$

Proof. From $z \in Z(p)$ we have $p \cdot z \geq p \cdot z'$; from $z' \in Z(p')$ we have $p' \cdot z' \geq p' \cdot z$. Adding yields $(p - p') \cdot (z - z') \geq 0$. □

Corollary (One-price changes, inputs vs. outputs). *If only component j of the price vector changes from p_j to p'_j with $p'_j > p_j$, then*

$$(z'_j - z_j)(p'_j - p_j) \geq 0.$$

Hence for an output j (quantities $z_j \geq 0$) we get $z'_j \geq z_j$ (more output when its price rises); for an input i (quantities $z_i \leq 0$) we get $z'_i \geq z_i$, i.e. the input is used less (moves toward zero).

“Afriat for firms”: rationalizing a finite dataset

Suppose we observe n price-choice pairs $\{(p^t, z^t)\}_{t=1}^n$.

Necessary condition (pairwise optimality). A necessary condition for profit maximization is

$$p^t \cdot z^t \geq p^t \cdot z^s \quad \text{for all } t, s \in \{1, \dots, n\}. \quad (\star)$$

Sufficient condition and a rationalizing technology. If $(\star)$ holds, then the closed, convex set

$$\mathcal{Z}^* := \{z \in \mathbb{R}^k : p^t \cdot z \leq p^t \cdot z^t \text{ for all } t\}$$

(with free disposal added if desired: $\mathcal{Z}^* + \mathbb{R}_-^k$) rationalizes the data: each z^t solves $\max_{z \in \mathcal{Z}^*} p^t \cdot z$. If, moreover, all observed profits are nonnegative ($p^t \cdot z^t \geq 0$ for all t), then $0 \in \mathcal{Z}^*$ (consistent with free disposal and the idea that “doing nothing” is feasible).

What we can and cannot infer. From finite data we cannot identify the true technology nor its curvature; we can only assert that there *exists* a closed, convex (free-disposal) technology that makes the observations profit maximizing.

Joaquín's comment: What Proposition 9.10 actually shows

Proposition 9.10 is slightly stronger than the exposition above. The inequalities

$$p^t \cdot z^t \geq p^t \cdot z^s \quad \forall t, s$$

are not only *necessary* but also *sufficient* for rationalizability ($\exists Z \subseteq \mathbb{R}^K : z^n \in \arg \max \{ p^n \cdot z : z \in Z \} \forall n$, then data set $\{(p^n, z^n)\}_{n=1}^N$ is rationalized by production set Z): they hold if and only if there exists a production set Z for which each observed production plan z^t maximizes profits at prices p^t . Nevertheless this is consistent with what is written in the notes. But I am trying to go beyond this.

The set

$$\mathcal{Z}^* = \{z \in \mathbb{R}^K : p^t \cdot z \leq p^t \cdot z^t, \forall t\}$$

is the *maximal* rationalizing technology. It is automatically closed and convex because it is the intersection of finitely many closed half-spaces, one generated by each observation. Thus, convexity is not assumed; it is a consequence of this construction.

Moreover, $\mathcal{Z}^*$ is not unique. Every production set Z satisfying

$$\{z^1, \dots, z^n\} \subseteq Z \subseteq \mathcal{Z}^*$$

also rationalizes the data. Hence, finite observations do not identify the firm's true technology, but only the family of production sets lying between the observed production plans and the maximal rationalizing technology.

